

N706-EA

Hardware User Guide

Cat.1 Module
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This document is specifically for N706-EA.

This document is intended for system engineers (SEs), development engineers, and test engineers.

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About This Document

Scope

This document is applicable to the N706-EA series. It describes N706-EA information and functional interface characteristics, and provides a reference design for each functional interface.

The reference design in this document is for reference only, and user applications must be designed according to actual scenarios and conditions. If you have any questions, please contact Neoway FAEs.

Audience

This document is intended for system engineers (SEs), development engineers, and test engineers.

Change History

Issue	Date	Change	Author
1.0	2023-12	Initial release.	Liang Hui
1.1	2024-01	Updated the supported sub-models.	Liang Hui
1.2	2024-08	- Updated the GNSS parameters. - Updated the ANT_GNSS reference design and the precautions.	Liang Hui
1.3	2024-08	- Updated the applicable region of the submodels N706-EA-02-S1 and N706-EA-12-S1 to Europe in Table 2-1. - Updated requirements for process moisture sensitivity and solder paste.	Liang Hui

Conventions

Symbol	Description
	Indicates danger or warning. This information must be followed. Otherwise, a catastrophic module or user device failure or bodily injury may occur.



Indicates caution. This symbol alerts the user to important points about using the module. If these points are not followed, the module or user device may fail.



Indicates instructions or tips. This symbol provides advices or suggestions that may be useful when using the module.

Related Documents

Neoway_N706-EA_Datasheet

Neoway_N706_Command_Mannual

Neoway_N706-EA_EVK_User_Guide

1 Safety Recommendations

Please read carefully and strictly adhere to the following safety recommendations to ensure that the product complies with national and environmental requirements, and to prevent threats to personal safety and potential damage to the product and workplace:

- Do not use in places where fire or explosion may occur.

Using the product in locations with flammable gases, dust, or substances like propane, gasoline, or flammable aerosols may lead to explosions or fires.

- In areas where wireless communication is prohibited, please disable the wireless communication function.

If the product is used in medical institutions or on airplanes, electromagnetic waves emitted by this product may interfere with surrounding equipment.

Please note the following requirements during the application design and use of this module product:

- Do not dismantle the product on your own, or you will lose after-sales warranty service.
- Design the product correctly according to the Hardware User Guide. Connect the product to a stable power voltage, and ensure wiring complies with safety and fire prevention requirements.
- Avoid touching the product pins to prevent static damage.
- Do not insert or remove USIM cards or mobile storage cards while the device is not turned off.

2 About N706-EA

N706-EA is an LTE industrial-grade cellular module developed based on Unisoc UIS8850, providing connectivity on FDD-LTE networks. With dimensions of (17.7 ± 0.10) mm $\times$ (15.8 ± 0.10) mm $\times$ (2.3 ± 0.15) mm, the module supports rich hardware interfaces including audio, video, WiFi and optional GNSS function. It is suitable for developing IoT communication devices such as wireless meter reading terminals, handheld POS, industrial routers and so on.

N706-EA has the following features:

- Cortex-A5 processor, 500 MHz CPU clock speed, 32 KB I-cache and 32 KB D-cache.
- Supported network mode: LTE Cat.1
- Supported functional interfaces: PCM, USIM, UART, I2C, USB2.0, SPI, and ADC.

2.1 Product Overview

The following lists the frequency bands and memory information that N706-EA supports.

Table 2-1 Supported frequency bands

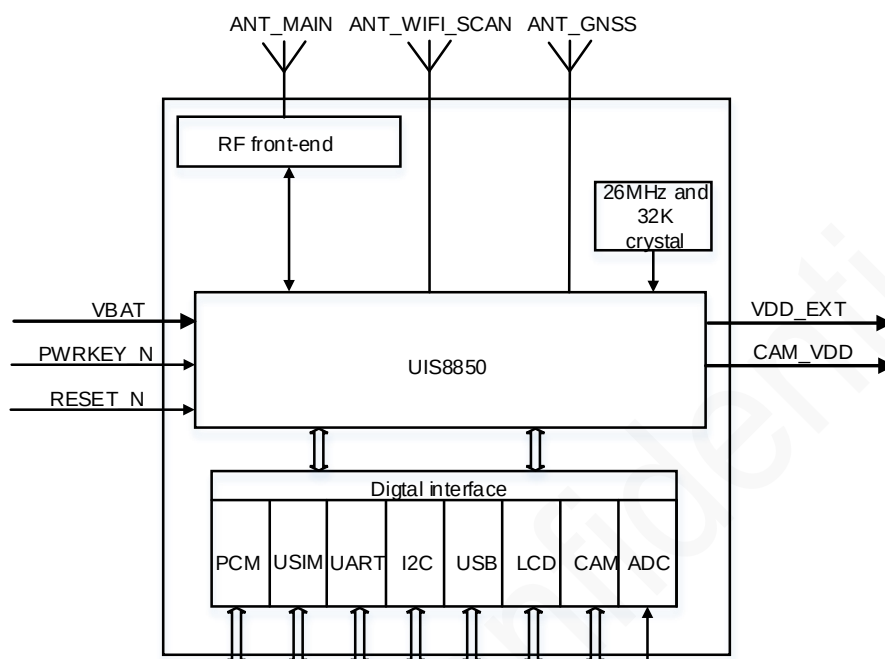
Variant	Region	Band	GNSS	WiFi-SCAN	Memory
N706-EA-01-P1	Europe	FDD-LTE: B1, B3, B5, B7, B8, B20, B28	Not supported	Supported	RAM: 8 MB ROM: 8 MB
N706-EA-02-S1	Europe	FDD-LTE: B1, B3, B5, B7, B8, B20, B28	Not supported	Supported	RAM: 8 MB ROM: 4 MB
N706-EA-12-S1	Europe	FDD-LTE: B1, B3, B5, B7, B8, B20, B28	Supported	Supported	RAM: 8 MB ROM: 4 MB

2.2 Block Diagram

The N706-EA module includes the following functional units:

- Baseband
- Digital interface (PCM, USIM, UART, I2C, USB 2.0, SPI)
- Analog interface (ADC)

Figure 2-1 Block diagram



2.3 Basic Features

Features	Description
Physical features	- Dimensions: (17.7±0.10) mm × (15.8±0.10) mm × (2.3±0.15) mm - Package: 109-pin LGA - Weight: TBD
Temperature ranges	Operating: -30°C to +75°C Extended*: -40°C* to +85°C* Storage: -40°C to +90°C
Operating voltage	VBAT: 3.4 V to 4.2 V, typical value: 3.8 V
Operating current	Sleep mode ¹⁾ : <1.8 mA Idle mode ²⁾ : <11 mA

Current in sleep mode¹⁾: means the current drawn by the module in sleep mode, a low power consumption state, in which its RF function is functioning properly but its peripheral interfaces are disabled. If there is an incoming call or SMS, the module will exit from the sleep mode, and after the incoming call or voice instant messaging has ended, the module will re-enter the sleep mode.

Current in standby mode²⁾: means the current drawn by the module in a normal operating mode, but no data service is being processed.

	Operating mode ³⁾ : <600 mA
Application processor	ARM Cortex- A5 processor, main frequency 500 MHz.
Band	FDD-LTE: B1, B3, B5, B7, B8, B20, B28
Wireless rate	FDD-LTE: Cat. 1, Max 10 Mbps (DL)/Max 5 Mbps (UL)
Application Interfaces	4G antenna, GNSS antenna, WIFI_SCAN antenna. The characteristic impedance of each antenna is 50 Ω .
	Three UART interfaces, auto-adaptive default baud rate up to 115200 bps. MAIN_UART for AT communications. AUX_UART for CP-side log capturing. DBG_UART for AP-side log capturing.
	Two USIM interfaces, supporting 1.8 V/3 V USIM cards.
	One USB2.0 interface, slave mode only.
	Three SPI interfaces One flash SPI; one LCD-dedicated SPI interface; one camera-dedicated SPI interface.
	Two 12-bit ADC interfaces, voltage detection range: 0.1 V - VBAT.
	Two I2C interfaces, for master mode only.
	One PCM interface
AT commands	3GPP Release 13 Neoway extended AT commands
SMS	PDU, TXT
Data	PPP, RNDIS, ECM
Protocol	TCP, UDP, MQTT, FTP, HTTP/HTTPS, SSL, TLS
Certification approval	CCC*, SRRC*, RoHS*, CTA*

* means under development

Current in operating mode³⁾: means the current drawn by the module when there is data communications.

3 Reference Standard

N706-EA is designed by referring to the following standards:

- 3GPP TS 36.521-1 V16.2.1 User Equipment (UE) conformance specification; Radio transmission and reception; Part 1: Conformance Testing
- 3GPP TS 36.124 V16.1.0 ElectroMagnetic Compatibility (EMC) requirements for mobile terminals and ancillary equipment
- 3GPP TS 21.111 V16.0.0 USIM and IC card requirements
- 3GPP TS 51.011 V4.15.0 Specification of the Subscriber Identity Module -Mobile Equipment (SIM-ME) interface
- 3GPP TS 31.102 V17.0.0 Characteristics of the Universal Subscriber Identity Module (USIM) application
- 3GPP TS 31.111 V17.0.0 Universal Subscriber Identity Module (USIM) Application Toolkit (USAT)
- 3GPP TS 27.007 V17.0.0 AT command set for User Equipment (UE)
- 3GPP TS 27.005 V16.0.0 Use of Data Terminal Equipment – Data Circuit terminating Equipment (DTE - DCE) interface for Short Message Service (SMS) and Cell Broadcast Service (CBS)

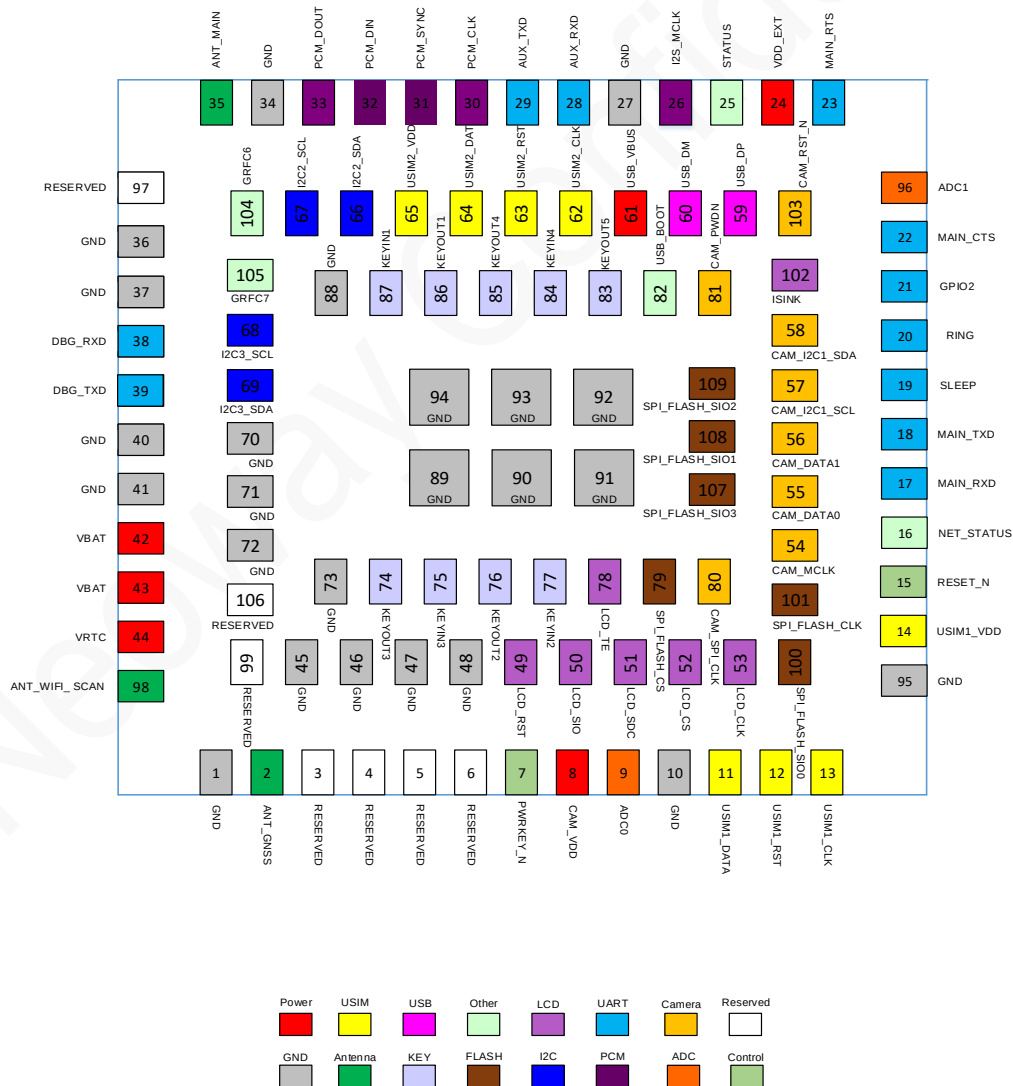
4 Pad Layout

N706-EA series modules are equipped with 109 pads, which are introduced in LGA package.

4.1 Pad Layout

The following figure shows the pad layout of N706-EA.

Figure 4-1 N706-EA pad layout (top view)



4.2 Pin Description

The following table lists the IO types and DC characteristics.

Table 4-1 IO types and DC characteristics

IO type		
B	Digital input and output, CMOS logic level	
DO	Digital output, CMOS logic level	
DI	Digital input, CMOS logic level	
PO	Power output	
PI	Power input	
AO	Analog output	
AI	Analog input	
AIO	Analog input and output	
DC characteristics		
Interface type	Power domain	Logic level characteristics
USIM	P1: Both 1.8 V and 3.0 V SIM types are supported. Activation and deactivation with an automatic voltage switch from 1.8 V to 3 V is implemented	1.8 V DC characteristics: $V_{IH}=0.7 \times V_{DD_P1} \sim V_{DD_P1}$ $V_{IL}=0V \sim 0.3 \times V_{DD_P1}$ $V_{OH}=0.9 \times V_{DD_P1} \sim V_{DD_P1}$ $V_{OL}=0V \sim 0.1 \times V_{DD_P1}$
		3 V DC characteristics: $V_{IH}=0.7 \times V_{DD_P1} \sim V_{DD_P1}$ $V_{IL}=0V \sim 0.3 \times V_{DD_P1}$ $V_{OH}=0.9 \times V_{DD_P1} \sim V_{DD_P1}$ $V_{OL}=0V \sim 0.1 \times V_{DD_P1}$
GPIO	P2:1.8V	$V_{IH}=0.7 \times V_{DD_P2} \sim V_{DD_P2}$ $V_{IL}=0V \sim 0.3 \times V_{DD_P2}$ $V_{OH}=0.9 \times V_{DD_P2} \sim V_{DD_P2}$ $V_{OL}=0V \sim 0.1 \times V_{DD_P2}$

Table 4-2 Pin description

Signal	Pin SN	I/O type	Function description	Level feature	Remarks
Power interfaces					
VBAT	42, 43	PI	Main power supply input	$V_{\min}=3.4\text{ V}$ $V_{\text{norm}}=3.8\text{ V}$ $V_{\max}=4.2\text{ V}$	The external power supply must ensure at least 2 A current.
VDD_EXT	24	PO	1.8 V power output	$V_{\text{norm}}=1.8\text{ V}$ $I_{\max}=50\text{ mA}$	IO interface, only used for voltage-level translation. Leave this pin open if unused.
VRTC	44	PI	RTC power supply	2.8 V - 3.2 V, typical value: 3.0 V	Supplies power for the real-time clock inside the module. It can be externally connected to button cells or Farah capacitors.
GND	1, 10, 27, 34, 36, 37, 40, 41, 45, 46, 47, 48, 70, 71, 72, 73, 88, 89, 90, 91, 92, 93, 94, 95				Make sure all GND pins are grounded.
Control interfaces					
PWRKEY_N	7	DI	Module on/off control	-	Active low This pin is connected to VBAT through a 20 k Ω pull-up resistor inside the module. For details, see section 5.2.1 .
RESET_N	15	DI	Module reset input	-	Active low This pin is connected to VBAT through a 20 k Ω pull-up resistor inside the module.
SLEEP	19	DI	SLEEP pin		

UART interface

MAIN_RXD	17	DI	Data input	P2	Software version (standard): For AT communication Open version: used for data transmission.
MAIN_TXD	18	DO	Data output	P2	
MAIN_CTS	22	DI	UART clear to send	P2	
MAIN_RTS	23	DO	UART request to send	P2	
AUX_RXD	28	DI	UART data input	P2	AUX _UART for CP-side log capturing.
AUX_TXD	29	DO	UART data output	P2	
DBG_RXD	38	DI	UART data input	P2	DBG_UART is only used for Debug. Can be used to capture AP log.
DBG_TXD	39	DO	UART data output	P2	

USIM interface

USIM1_DATA	11	B	USIM1 data	P1	Connecting this pin to USIM1_VCC through a 4.7 kΩ pull-up resistor is required.
USIM1_RST	12	DO	USIM1 reset	P1	-
USIM1_CLK	13	DO	USIM1 clock	P1	-
USIM1_VDD	14	PO	USIM1 power	P1	-
USIM2_VDD	65	PO	USIM2 power	P1	-
USIM2_RST	63	DO	USIM2 reset	P1	-
USIM2_CLK	62	DO	USIM2 clock	P1	-
USIM2_DATA	64	B	USIM2 data	P1	Connecting this pin to USIM2_VCC through a 4.7 kΩ pull-up resistor is required.

USB interface

USB_VBUS	61	PI	USB power supply	3.5 V - 5.25 V, typical value: 5 V	-
USB_DP	59	AIO	USB data +	-	Route the DM and DP traces as differential pairs with 90 Ω impedance.
USB_DM	60	AIO	USB data -	-	

ADC interface

ADC0	9	AI	General A/D interface	-	12-bit, detectable voltage range: 0.1 V - VBAT. Leave this pin open if unused.
ADC1	96	AI	General A/D interface	-	12-bit, detectable voltage range: 0.1 V - VBAT. Leave this pin open if unused.

I2C interface

I2C2_SDA	66	B	I2C data	P2	Pulling this pin up to VDD_EXT is required.
I2C2_SCL	67	DO	I2C clock	P2	Pulling this pin up to VDD_EXT is required.
I2C3_SDA	69	B	I2C data	P2	- Pulling this pin up to VDD_EXT is required. - Note: This pin may exhibit a transient, undefined state at the millisecond level when the module undergoes a power-down process. If you have the requirement to repurpose this pin as a General-Purpose Input/Output (GPIO), it is
I2C3_SCL	68	DO	I2C clock	P2	

recommended to engage with Neoway FAEs to implement an RC filter circuit.

PCM interface

PCM_CLK	30	DO	PCM data clock	P2	Can be multiplexed into I2S_BCK.
PCM_SYNC	31	DO	PCM data frame synchronization	P2	Can be multiplexed into I2S_LRCK.
PCM_DIN	32	DI	PCM data input	P2	Can be multiplexed into I2S_SDAT_I.
PCM_DOUT	33	DO	PCM data output	P2	Can be multiplexed into I2S_SDAT_O.
I2S_MCLK	26	DO	Data clock	P2	

LCD interface

LCD_SDC	51	DO	SPI data or command control	P2	-
LCD_SIO	50	B	SPI data input/output	P2	-
LCD_CS	52	DO	SPI chip select signal of the slave device	P2	-
LCD_CLK	53	DO	SPI clock	P2	-
LCD_RST	49	DO	LCD reset	P2	-
LCD_TE	78	DO	LCD data frame synchronization	P2	-
ISINK	102	PI	LCD sink current input pin, used to adjust backlight brightness.		Connect this pin to the cathode of an LCD and its brightness is adjusted by a sink current. I _{max} = 81 mA.

FLASH interface

SPI_FLASH_CS	79	DO	SPI FLASH chip select	P2	- Note: This pin may exhibit a transient, undefined state at the millisecond level when the module undergoes a power-down process. If you have the requirement to repurpose this pin as a General-Purpose Input/Output (GPIO), it is recommended to engage with Neoway FAEs to implement an RC filter circuit. - Pin79 can be multiplexed into USIM1_DET.
SPI_FLASH_CLK	101	DO	SPI FLASH clock	P2	
SPI_FLASH_SIO0	100	B	SPI FLASH data 0	P2	
SPI_FLASH_SIO1	108	B	SPI FLASH data 1	P2	
SPI_FLASH_SIO2	109	B	SPI FLASH data 2	P2	
SPI_FLASH_SIO3	107	B	SPI FLASH data 3	P2	

CAM interface

CAM_VDD	8	PO	CAM analog power	$V_{min}=1.61\text{ V}$ $V_{norm}=1.8\text{ V}$ $V_{max}=3.2\text{ V}$	This analog power is disabled by default. If enabled, the default output is 1.8 V / 100 mA.
CAM_PWDN	81	DO	CAM power-down control	P2	Leave this pin open if unused.
CAM_DATA0	55	B	CAM SPI data 0	P2	Camera-dedicated SPI interface
CAM_DATA1	56	B	CAM SPI data 1	P2	
CAM_SPI_CLK	80	DO	CAM SPI clock	P2	
CAM_RST_N	103	DO	CAM reset	P2	-
CAM_MCLK	54	DO	CAM clock signal	P2	-
CAM_I2C1_SCL	57	DO	CAM I2C clock		
CAM_I2C1_SDA	58	B	CAM I2C data		

Antenna interface					
ANT_GNSS	2	AI	GNSS antenna	-	50 Ω impedance.
ANT_WIFI_SCAN	98	AI	WIFI_SCAN antenna	-	50 Ω impedance.
ANT_MAIN	35	AIO	Main antenna	-	50 Ω impedance.
Matrix keypad interface					
KEYOUT3	74	DO	Matrix keypad output 5	P2	1.8 V voltage domain. Leave these pins unconnected if unused. External pull-up/ pull-down is prohibited. If this pin needs to be multiplexed as a GPIO, please contact Neoway FAEs to confirm the initial state of the GPIO.
KEYIN3	75	DI	Matrix keypad input 5	P2	
KEYOUT2	76	DO	Matrix keypad output 2	P2	
KEYIN2	77	DI	Matrix keypad input 2	P2	
KEYOUT5	83	DO	Matrix keypad output 4	P2	
KEYIN4	84	DI	Matrix keypad input 6	P2	
KEYOUT4	85	DO	Matrix keypad output 6	P2	
KEYOUT1	86	DO	Matrix keypad output 1	P2	
KEYIN1	87	DI	Matrix keypad input 1	P2	
Other functional interfaces					
STATUS	25	DO	Operating status indication	P2	Leave this pin open if unused.
NET_STATUS	16	DO	Network status indication	P2	Leave this pin open if unused.
USB_BOOT	82	DI	Emergency download control	P2	Ground this pin and then power up the module to enter the USB forced download mode.
GRFC6	104	DO	General RF control IO6	P2	General RF control IO
GRFC7	105	DO	General RF control IO7	P2	Leave this pin open if unused.

RESERVED	3, 4, 5, 6, 97, 99, 106	Reserved pin	-	Reserved for future function expansion or functions not open to the user. These RESERVE pins might have different functions. Leave these RESERVED pins open.
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5 Application Interfaces

N706-EA provides the control, communications, peripherals, display, and RF interfaces to meet the functional requirements of customers in different application scenarios.

This chapter describes how to design each interface and provides reference designs and guidelines.

5.1 Power Interfaces

The schematic design and PCB layout of the power supply part are the most critical process in application design and they will determine the performance of customers' applications. Please read the design guidelines of power supply and comply with the correct design principles to obtain the optimal circuit performance.

Signal	Pin SN	I/O	Function description	Remarks
VBAT	42, 43	PI	Main power supply of the module	$V_{\min}=3.4\text{ V}$ $V_{\text{norm}}=3.8\text{ V}$ $V_{\max}=4.2\text{ V}$
VDD_EXT	24	PO	1.8 V power output	$V_{\text{norm}}=1.8\text{ V}$ $I_{\max}=50\text{ mA}$
VRTC	44	PI	RTC power supply	2.8 V - 3.2 V, typical value: 3.0 V
GND	1, 10, 27, 34, 36, 37, 40, 41, 45, 46, 47, 48, 70, 71, 72, 73, 88, 89, 90, 91, 92, 93, 94, 95			Make sure all GND pins are grounded.

5.1.1 VBAT

The power supply design encompasses two key facets: schematic design and PCB layout. These are integral in sustaining the optimal functionality and safety of the module.

Schematic Design of Power Supply



The module may experience a momentary peak current of up to 2 A during network registration. It is necessary to parallel a large capacitor on VBAT to enhance the current sustaining ability. This is to ensure that the VBAT input voltage does not fall below the module's operating voltage during these moments of peak current.

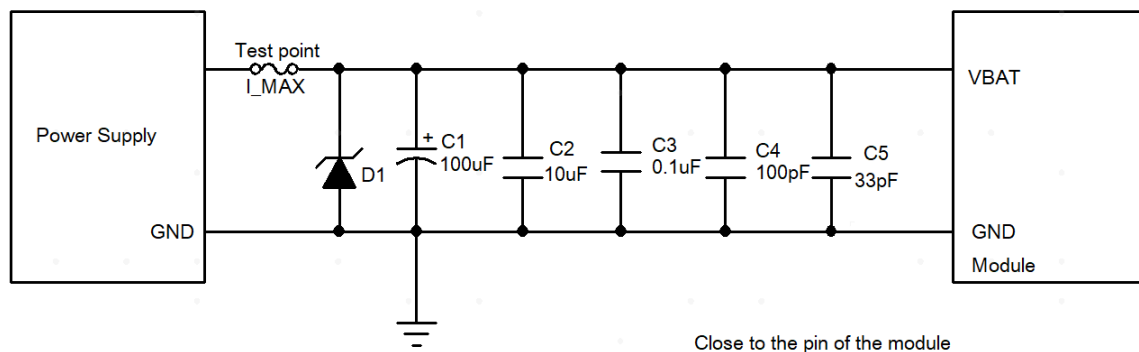
Never bridge a voltage drop between a higher input and the module power supply with a diode. Diodes' forward voltage drop V_f can cause instability or damage, particularly at high currents or low temperatures.

The power supply design of the N706-EA module depends on the input voltage, categorized into:

- 3.4 V - 4.2 V power input (typical value: 3.8 V, battery supplied)
- 4.2 V - 5.5 V power input (typical value: 5.0 V, internal computer rectifier)
- 5.5 V - 24 V power input (typical value: 12 V, system DC power in industries like automotive)

Design Recommendations for 3.4 V - 4.2 V Input:

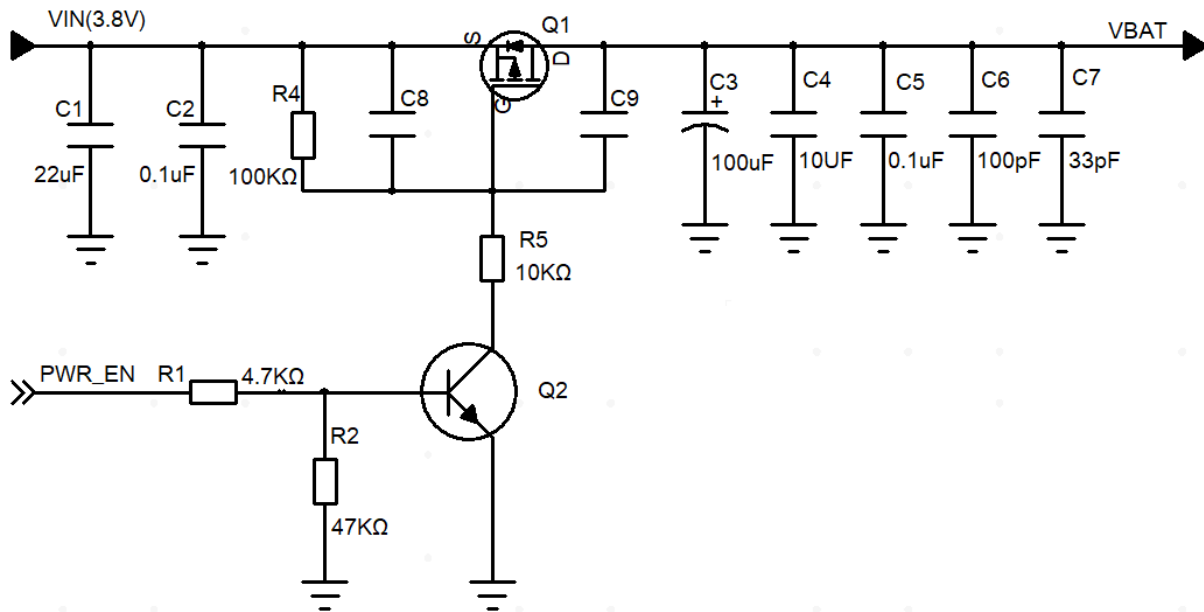
Figure 5-1 Recommended design of the power supply circuit 1



- The maximum input voltage of the module power supply is 4.2 V, and the typical value is 3.8 V.
- In order to get a stable power source, it is recommended to select a TVS diode with suggested low reverse voltage (V_{RWM} 4.5 V) and peak power ($P_{pp} = 2800$ W ($t_p = 8/20$ us)) at D1. Keep the placement of the TVS diode close to the power input interface to suppress surge voltage before it enters back-end circuits to prevent the back-end component and the module from damaging.
- Use a large bypass tantalum capacitor (220 μ F or 100 μ F) or an aluminum capacitor (470 μ F or 1000 μ F) at C1 to decrease voltage drops during bursts. Its maximum safe operating voltage should be larger than 2 times the voltage of the power supply.
- Keep the placement of the low-ESR bypass capacitors (C2, C3, C4, C5) close to the module to filter out high-frequency jamming from the power supply.

If it is necessary to control the power supply, the following circuit design is recommended:

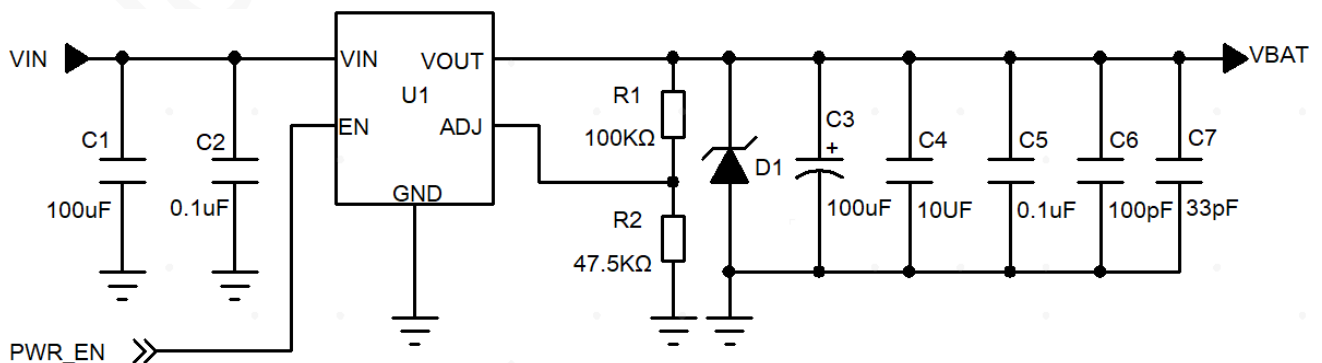
Figure 5-2 Recommended design of the power supply circuit 2



- Utilize an enhanced p-MOSFET at Q1, of which the safe operating voltage is at least 12 V ($V_{ds} = -12\text{ V}$) and drain current is at least 3.5 A ($I_{D(MAX)} = -3.5\text{ A}$) and R_{ds} is low ($R_{ds(on)} = 108\text{ m}\Omega$).
- Select a common NPN transistor at Q2. Reserve enough tolerances for R1 and R2, ensuring that R2 is at least 10 times greater than R1, as the break-over voltage on the transistor's base increases at low temperatures.
- Keep the placement of C3 close to the module. Select a tantalum electrolytic capacitor with large capacitance (220 μF or 100 μF) or an aluminum electrolytic capacitor (470 μF or 1000 μF) at C3 to improve the instantaneous large freewheeling current of the power supply, and its withstand voltage value should be larger than 2 times the power supply voltage.
- Keep the placement of the low-ESR ($\leq 0.7\Omega$) bypass capacitors (C4, C5, C6, C7) close to the module to filter out high-frequency jamming from the power supply.

The design recommendations for the 4.2 V - 5.5 V power input are as follows:

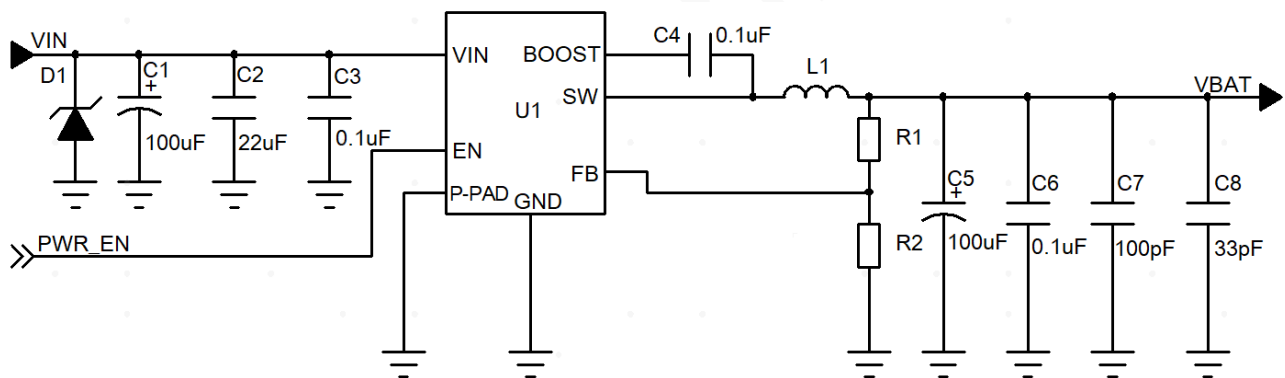
Figure 5-3 Recommended design of the power supply circuit 3



- Design with LDO is simpler and more efficient when the output of power supply is close to the permissible voltage across VBAT.
- Select an LDO with maximum output current above 2 A at U1 to ensure the normal performance of the module.
- In order to get a stable power source, it is recommended to use a TVS diode with suggested low reverse voltage ($V_{RWM} 4.5\text{ V}$) and peak power ($P_{pp} = 2800\text{ W}$ ($t_p = 8/20\text{ us}$)) at D3. Keep the placement of the TVS diode close to the power input interface to suppress surge voltage before it enters back-end circuits to prevent the back-end component and the module from damaging.
- Keep the placement of C3 close to the module. Select a large tantalum electrolytic capacitor ($220\text{ }\mu\text{F}$ or $100\text{ }\mu\text{F}$) or aluminum electrolytic capacitor ($470\text{ }\mu\text{F}$ or $1000\text{ }\mu\text{F}$) at C3 to improve the instantaneous large freewheeling current of the power supply, and its withstand voltage value should be larger than 2 times the power supply voltage.
- Keep the placement of the low-ESR bypass capacitors (C4, C5, C6, C7) with capacitance not greater than $0.7\text{ }\Omega$ close to the module to filter out high-frequency jamming from the power supply.

The design recommendations for the 5.5 V - 24 V power input are as follows:

Figure 5-4 Recommended design of the power supply circuit 4



- If there is a high voltage drop between the power input and VBAT, a DC-DC design should be selected for higher efficiency, and the output current should be at least 2 A.
- Keep the TVS diode (D1) close to the power input interface to ensure that the power surge voltage is clamped before entering the back-end circuit, thus protecting the back-end components and modules.
- Keep the placement of C5 close to the module. Select a large tantalum electrolytic capacitor ($220\text{ }\mu\text{F}$ or $100\text{ }\mu\text{F}$) or aluminum electrolytic capacitor ($470\text{ }\mu\text{F}$ or $1000\text{ }\mu\text{F}$) at C3 to improve the instantaneous large freewheeling current of the power supply, and its withstand voltage value should be larger than 2 times the power supply voltage.
- Keep the low ESR ($\leq 0.7\text{ }\Omega$) bypass capacitors (C6, C7, C8) as close to the module as possible to filter out high-frequency interference in the power supply.



When the module fails to be switched off normally due to abnormality, it is recommended to cut off the power first to shut down the module, and then power on again to restart the module.

PCB Layout Guidelines

Place an ESR capacitor at the output of the power source to suppress the peak current. Place TVS diodes at the input end of the module to suppress voltage spikes and protect back-end devices.

Several key points in power supply design are summarized below:

- The TVS diode can absorb instantaneous high-power pulse, and withstand instantaneous pulse current peaks up to tens or even hundreds of amperes, with extremely short response time of voltage clamping. Place the TVS diode near the power input. This helps control surge voltage and prevents pulses from affecting nearby PCB wires.
- Place bypass capacitors as close as possible to the power supply interface of the module to filter out high-frequency noise signals in the power supply.
- For the main power circuit of the module, ensure that the PCB trace is wide enough that 2 A current can be safely passed, with no significant loop voltage drop. Ensure that the ground plane of the power supply part is as complete as possible. In addition, keep the power cable short and thick.
- Keep noise-sensitive circuits, such as audio circuits and RF circuits far away from the power circuitry, especially when the DC-DC power supply is used.
- The voltage frequency of the SW pin of the DC-DC power supply is high, so the loop area should be minimized. Keep sensitive components far away from the SW pin of the DC-DC component to prevent noise coupling. Place feedback components as close as possible to the FB pin.
- The thermal-dissipation pads and GND pads of the DC-DC chip should be grounded to ensure good thermal dissipation and noise signal isolation.

5.1.2 VDD_EXT



It is recommended to use VDD_EXT only for voltage-level translation and an ESD protector should be added.

N706-EA provides one VDD_EXT interface that supports 1.8 V voltage and up to 50 mA current. There is no output voltage at this pin in PSM mode.

5.1.3 VRTC

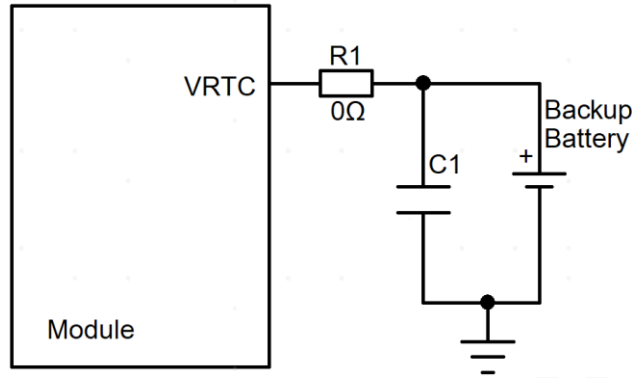
VRTC is internally connected to the real-time clock of the module. The VRTC pin can be connected to a button battery or farrar capacitor to supply power for the real-time clock.

When the module is in normal operating mode (with a valid voltage applying at the VBAT), the typical

output voltage of VRTC is 3.0 V and the maximum current is 2 mA. After the VRTC pin is connected to a button battery or farrar capacotor, the RTC inside the module can maintain a normal operation for a short time if the voltage at VBAT is disconnected. Leave this pin open if unused.

The following shows the reference design of the VRTC power supply.

Figure 5-5 Reference design of the VRTC power supply



5.2 Control Interfaces

Signal	Pin SN	I/O	Function description	Remarks
RESET_N	15	DI	Module reset input	Active low
PWRKEY_N	7	DI	Module on/off control	Active low

5.2.1 Module Power-on

Table 5-1 Power-on description

Notice Power-on trigger method	Supported or not	Power-off method	Notice
Button	Supported	Press the power-on button for greater than 3s and then release it to switch off the module.	For example: Do not disconnect the power supply directly when the module is in power-on mode, as this may damage the internal flash of the module.
Pulse	Supported	A low pulse input (controlled by MCU) for greater than 3s can shut down the module.	

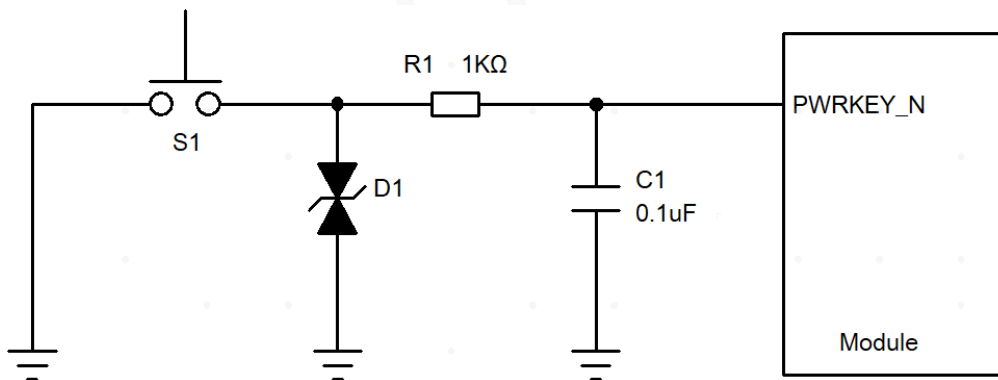
Auto power-on	Supported	Send the power-off AT command to the module and then disconnect its power source.	For example: When using AT commands to shutdown the module, if the system is not powered down (power-down means not-powered mode), the module may start again after it is shut down.
USB startup	Supported	Disconnect the USB, send the power-off AT command to the module, and then power down it.	For example: When using AT commands to switch off the module, if the system is not powered down (not-powered mode) and the USB is still connected, the module may start again after power-off.

Power-on controlled by PWRKEY_N

The following show two recommended reference designs of startup controlled:

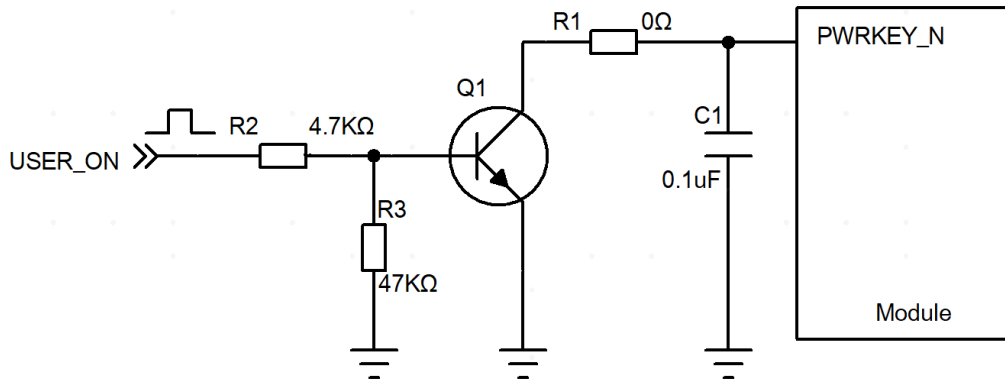
- Figure 5-6 shows the reference design of key-control startup.

Figure 5-6 Reference design for push-button startup



- Figure 5-7 shows the reference design of module power-on through an MCU.

Figure 5-7 Reference design for MCU-controlled startup

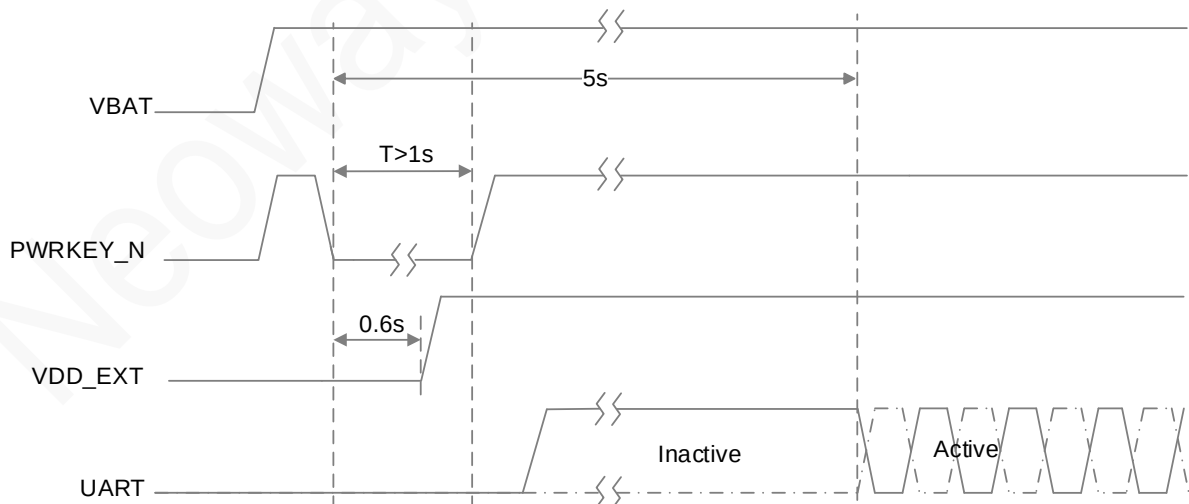


Power-on Process

After VBAT is powered on, releasing the PWRKEY_N pin after inputting a low-level pulse of more than 1s can trigger the module to start up, and it can be controlled by AT commands normally after 5s. It is important not to connect an external resistor with high resistance in series with the PWRKEY_N pin, as this could keep the pin at a high level and prevent the module from starting. It is recommended to connect a 1 kΩ resistor in series to the PWRKEY_N pin.

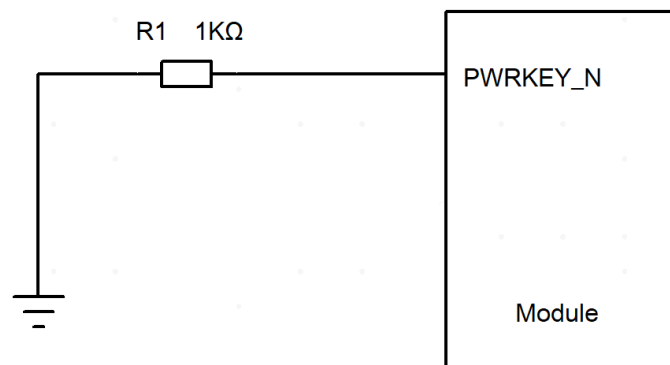
- Avoid performing operations on the module until initialization is complete, as the states of each pin are uncertain if the initialization process has not been finished. Refer to Figure 5-8 for the power-on process.

Figure 5-8 Power-on process



Automatic power-on once powered-up

Figure 5-9 Reference design of automatic startup once powered up



If you implement the automatic startup function but the shutdown function is not required, you can directly pull up PWRKEY_N to ground. The recommended pull-down resistance is 1 kΩ.

5.2.2 Module Power-off



When the module is in power-on mode, do not directly disconnect the power supply of the module; otherwise, the flash memory (Flash) inside the module may be damaged. It is recommended that before disconnecting the power supply, you should switch off the module through PWRKEY_N or a dedicated AT command.

When shutting down using an AT command, ensure that PWRKEY_N is not in a low-level state after executing the shutdown command. If it is, the module will restart after completing the shutdown process without the power being disconnected.

Power-off Methods:

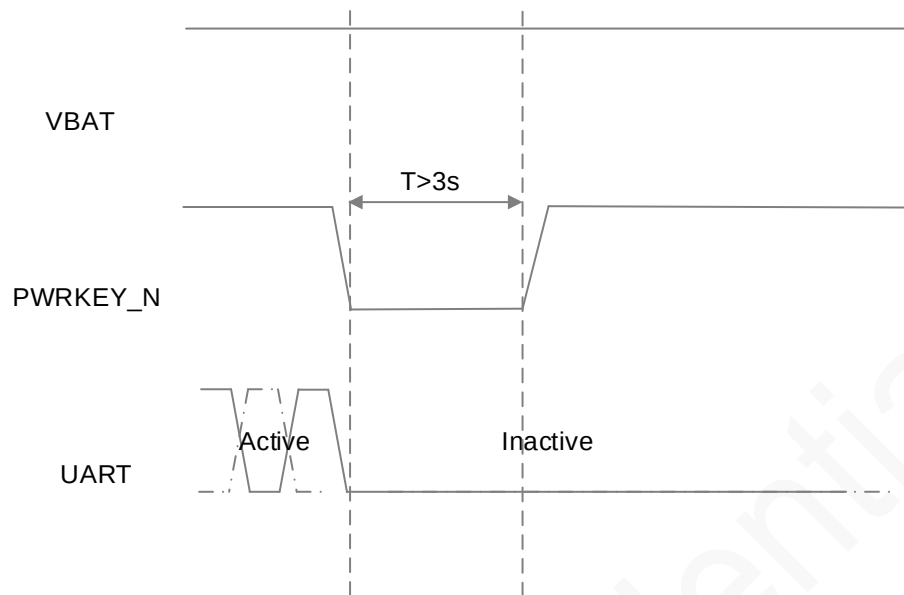
- Hard power-off: power-off controlled by PWRKEY_N
- Software Power-off via AT command

Power-off controlled by PWRKEY_N

When the module is in power-on mode, to turn off the module, the PWRKEY_N pin must be asserted low for greater than 3s and then released.

The following figure shows the hardware power-off process:

Figure 5-10 N706-EA hardware power-off process



Power-off controlled by AT command

When the module is in power-on mode, to turn off the module, send it the power-off AT command. Whether the module is switched off by hardware or software method, their mechanisms are the same, that is, by applying a low level at the PWRKEY_N pin to switch off the module.

For details, see the *Neoway_N706_AT_Commands_Manual*.

5.2.3 Resetting Methods



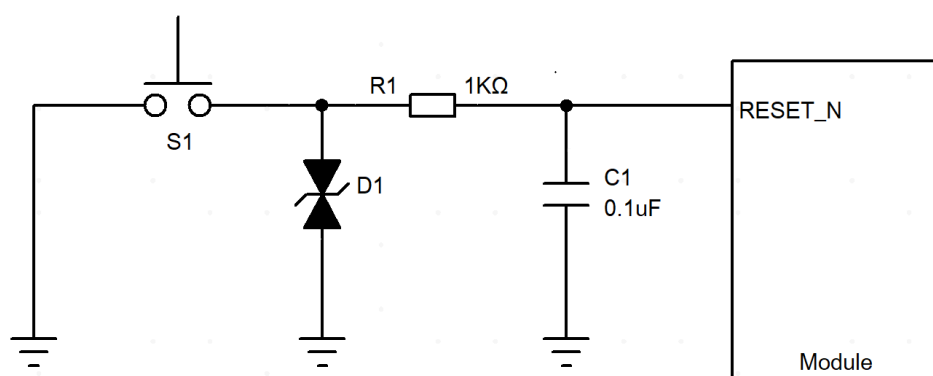
If the module experiences anomalies (such as freezing), a system reset can be performed through the RESET_N pin to ensure normal operation. Therefore, it is recommended to add a reset control circuit during circuit design, and the recommended reference circuit is shown in Figure 5-11 and Figure 5-12.

When the module is in power-on mode, inputting a low pulse at RESET_N for more than 100 ms can reset the module. For the module reset process, see Figure 5-11.

The following show two recommended reference designs of resetting N706-EA:

- Reference design for push-button reset

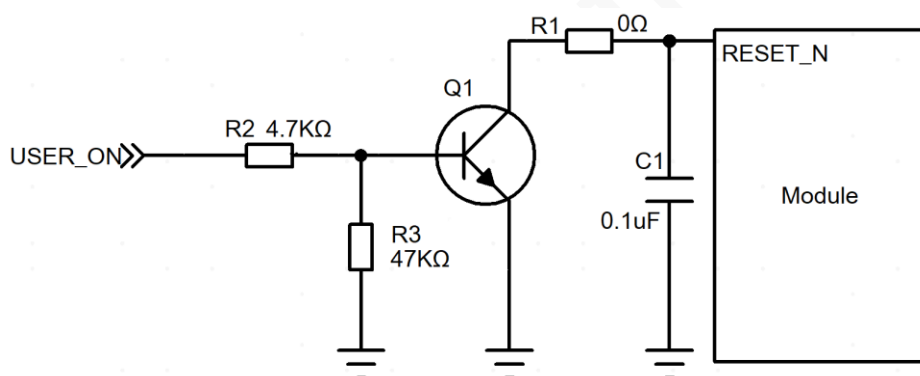
Figure 5-11 Reference design for push-button reset



- Reference design of MCU-controlled reset

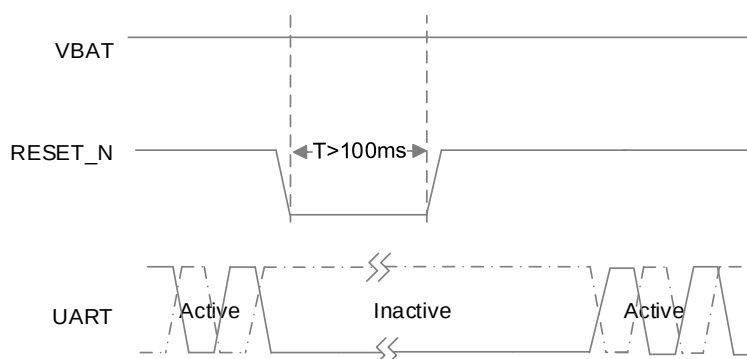
If you use a 1.8 V / 2.8 V / 3.0 V IO system, it is recommended to add a triode to for isolation. Refer to the following figure for specific design.

Figure 5-12 Reference reset circuit in which a triode is used for isolation



N706-EA module reset process is shown in the following figure.

Figure 5-13 Module reset process



5.2.4 SLEEP

Signal	Pin SN	I/O	Function description	Remarks
SLEEP	19	DI	Sleep and Wakeup	-

The SLEEP pin is used to control sleep mode of the module and needs0 to be used together with the AT command. Entering sleep mode or waking up the module by executing AT commands through the USB interface is supported. For details, see *Neoway_N706_AT_Commands_Manual*.

The following show the processes:

Figure 5-14 Process of entering into sleep mode

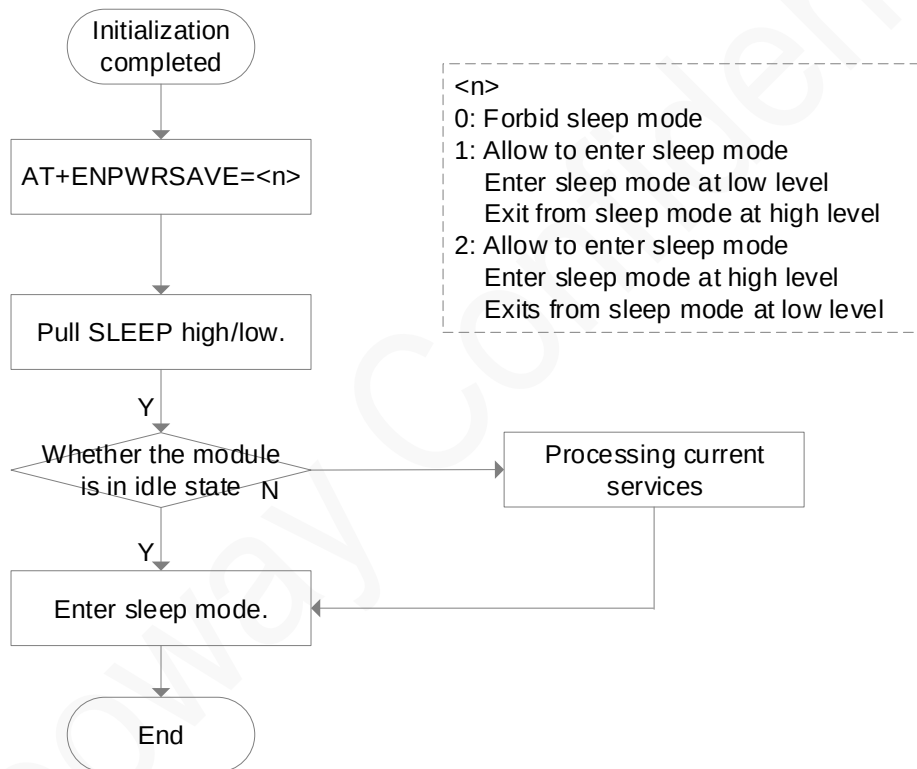


Figure 5-15 Process of processing data service in sleep mode

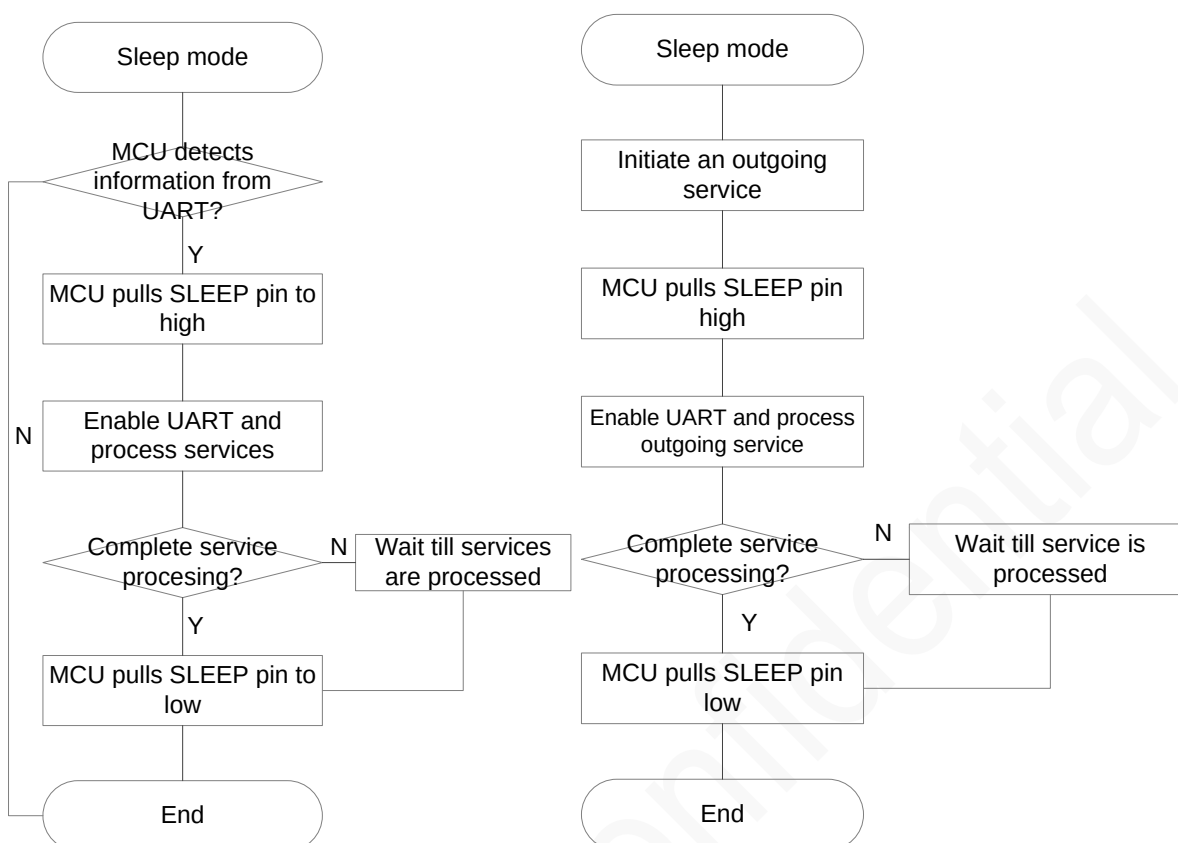
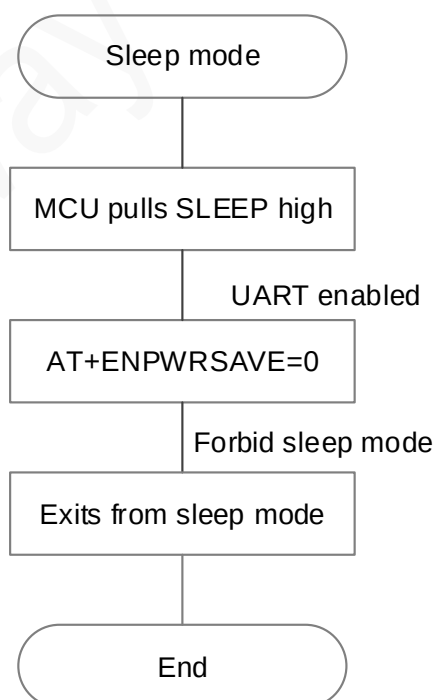


Figure 5-16 Process of existing from the sleep mode

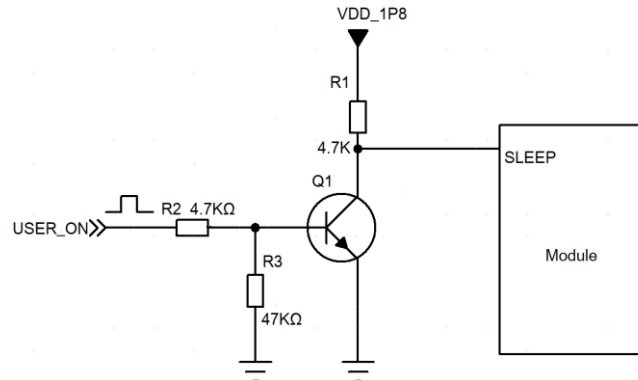


If you want to perform the wake-up function by using the sleep pin, it is recommended to design your circuit by referring to Figure 5-17.

Enter/exit sleep mode (send "AT+ENPWRSAVE=2" command to the serial interface of the module to enable sleep and wake-up function of the module.)

- Entering into sleep mode: keep USER_CTR low
- Exit from sleep mode: keep USER_CTR high

Figure 5-17 Reference design of module sleep and wake-up control



5.3 Peripheral Interfaces

N706-EA offers a variety of peripheral interfaces.

Naming Conventions:

- The I/O direction indicated by the module peripheral interface pin name is in line with the module.
- The peripheral pin naming follows the peripheral component itself. For example, UART_TXD is the module's data send pin and MCU_RXD is the MCU's receive. These pins should be interconnected.

In the process of MCU selection and design, it is important to note whether the pin signal naming is based on the module or the MCU.

5.3.1 USB

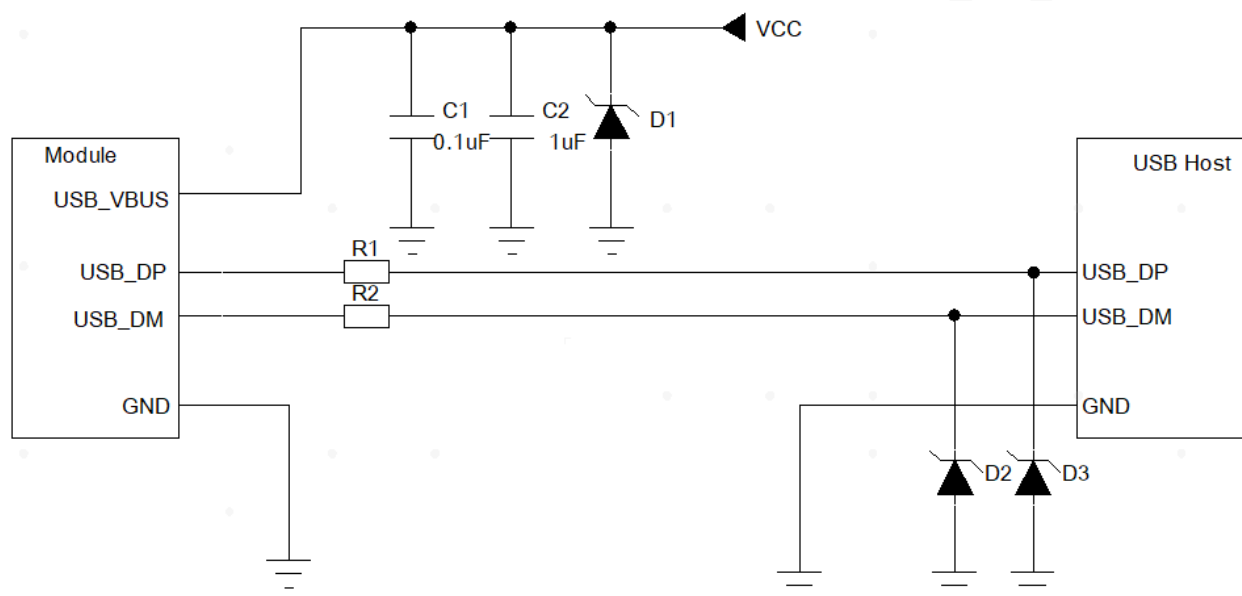


The USB interface must be reserved for software debug.

Signal	Pin SN	I/O	Function description	Remarks
USB_VBUS	61	PI	USB power supply	$3.5\text{ V} < \text{USB_VBUS} < 5.25\text{ V}$, typical value: 5 V.
USB_DP	59	AIO	USB data +	-
USB_DM	60	AIO	USB data -	-

N706-EA can implement program download, data communications, and debugging through the USB interface. Only slave mode is supported for USB of the module, which can be used as required. The recommended USB connection circuit is shown in Figure 5-18.

Figure 5-18 Recommended design of the USB interface



Schematic Design Guidelines

- Connect a 1 μF (C1) and a 33 pF (C2) filter capacitors in parallel to the USB_VBUS trace. An additional component (D1) is needed at the USB_VBUS pin as ESD protection and its reverse working voltage V_{RWM} should be greater than 6 V.
- Ensure that the junction capacitance of the ESD components (D2 and D3) in parallel to the USB_DP and USB_DM traces be less than 0.5 pF.
- Resistors of less than 10 Ω are needed in series at the USB_DM and USB_DP pins (R1, R2) to enhance ESD performance.

PCB Design Guidelines:

- Place the filter capacitor on the USB_VBUS trace as close to the module pins as possible, and place the ESD component as close to the USB connector as possible.

- Place the paralleled ESD components (D2, D3) on the USE_DP and USB_DM traces as close to the USB connector as possible.
- It is essential to route USB signal traces as differential pairs with ground surrounding. The impedance of the USB differential traces should be 90 Ω . Isolate the traces from the port to the module from other signal traces.

5.3.2 UART



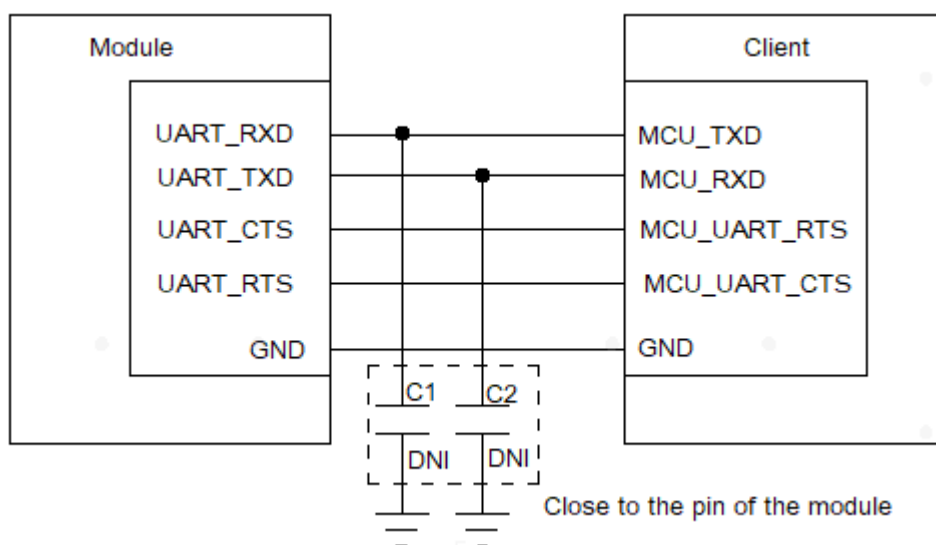
In environments with complex electromagnetic interference, the module's UART is susceptible to external electromagnetic interference, which can cause serious communication abnormalities. Therefore, please note the following when routing the UART signal trace on the PCB:

- Parallel a 5 to 10 pF capacitor near the module's UART_RXD and UART_TXD pins for filtering.
- Route UART traces far away from potential sources of interference.
- Route UART_RXD and UART_TXD traces on the inner layers of the PCB, maintain short trace lengths, avoid extensive routing on the top or bottom layers, and use a three-dimensional ground layout for these pins to enhance signal integrity.

Signal	Pin SN	I/O	Function description	Remarks
MAIN_RXD	17	DI	Data input	Software version (standard): For AT communication Open version: used for data transmission.
MAIN_TXD	18	DO	Data output	
MAIN_CTS	22	DI	Clear to send	
MAIN_RTS	23	DO	Request to send	
AUX_RXD	28	DI	UART data input	AUX_RXD can be used to capture CP-side log.
AUX_TXD	29	DO	UART data output	
DBG_RXD	38	DI	UART data input	DBG_RXD is use only for debugging. Can be used to capture AP log.
DBG_TXD	39	DO	UART data output	

N706-EA provides three UART interfaces. The UART supports a 1.8 V voltage and its default baud rate is 115200 bps. The following figure shows recommended design of the interface.

Figure 5-19 Reference design of the UART interface



Schematic Design Guidelines

- Ensure signal flow direction matches the connection orientation.
- It is prohibited to use diodes for voltage-level translation.
- Use capacitors C1 and C2, typically 5 - 10 pF, adjusting as necessary based on testing.

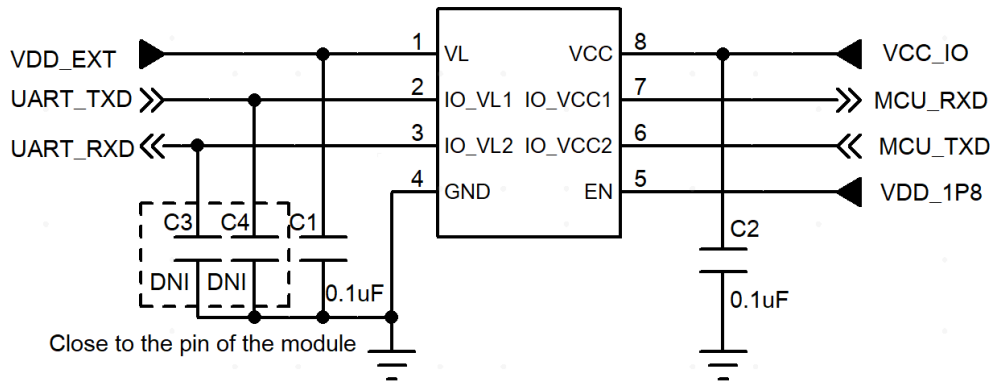
Select one of three recommended circuits based on logic level and rate differences.



Adjust component values in these circuits based on actual test outcomes. Pay attention to the different types of voltage-level translation circuits.

- If the UART baud rate is greater than 115200 bps, it is recommended to refer to the recommended voltage-level translation circuit 1.

Figure 5-20 Recommended voltage-level translation circuit 1

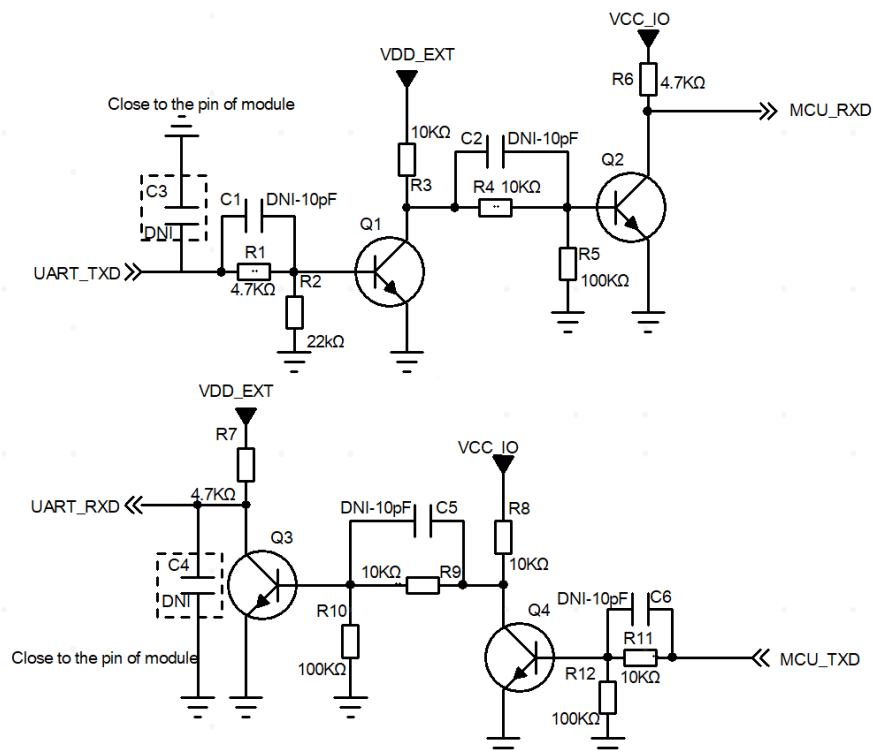


- VL: reference voltage for IO_VL1 and IO_VL2 (1.5 V - 5.5 V).
 - VCC: reference voltage for IO_VCC1 and IO_VCC2 (1.5 V - 5.5 V).
 - EN is the enable pin, which works at a voltage of greater than VL-0.2 V. In the above circuit, the EN pin is connected to VDD_EXT and the voltage translator is always working.
 - Use capacitors C3 and C4, typically 5 - 10 pF, adjusting as necessary based on testing.
- Dual-Triode Voltage-Level Translation Circuit (Baud Rate $\leq$ 115200 bps):
If the UART baud rate is less than or equal to 115200 bps, designing the TXD and RXD serial interfaces by referring to recommended voltage-level translation circuit 2 and designing the CTS and RTS serial interfaces by referring to recommended voltage-level translation circuit 3 are recommended. As shown in Figure 5-21 and Figure 5-22.



Adjust component values in these circuits based on actual test outcomes. Pay attention to the different types of voltage-level translation circuits.

Figure 5-21 Recommended voltage-level translation circuit 2



MCU_TXD and MCU_RXD: MCU's transmitting and receiving interfaces.

UART_TXD and UART_RXD: Module's transmitting and receiving interfaces. VCC_IO is the IO voltage of the MCU. VDD_EXT is the IO voltage of the module.

Schematic Design Guidelines

- Ensure the triode's base voltage operates within the specified temperature range for full conduction.
- Reserve acceleration capacitors (C1, C2, C5, C6) to potentially adjust the circuit's delay.
- Use capacitors C3 and C4, typically 5 - 10 pF, adjusting as necessary based on testing.

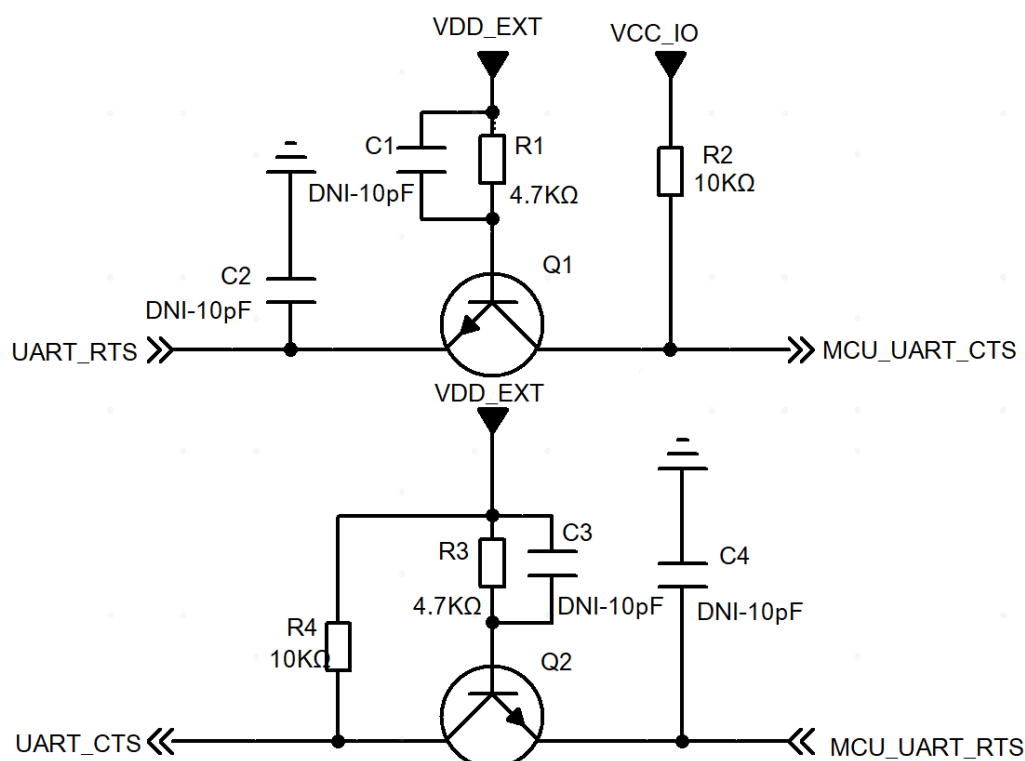
• Single-Triode Voltage-Level Translation Circuit:

The circuit translates the voltage through the turn-on and turn-off of the triode, and dual triode can achieve higher pressure difference after translation.

Schematic Design Guidelines

- Ensure the triode's base voltage operates within the specified temperature range for full conduction.
- It is recommended to reserve the acceleration capacitor, which can adjust the delay of the voltage-level translation circuit in some cases.

Figure 5-22 Recommended voltage-level translation circuit 3



MCU_CTS and MCU_RTS are the MCU-side signals, while UART_CTS and UART_RTS are the module-side signals. VCC_IO is the IO voltage of the MCU. VDD_EXT is the IO voltage of the module.

This circuit is a one-way translation solution, operating through the triode's conduction and cutoff. Please note the signal flow direction.

Schematic Design Guidelines

- Maintain the voltage difference between high and low levels below 2 V.
- It is recommended to reserve accelerating capacitors (C1 and C3) and adjust them according to test results.
- The base voltage of the triode is the lower voltage of the two sides.

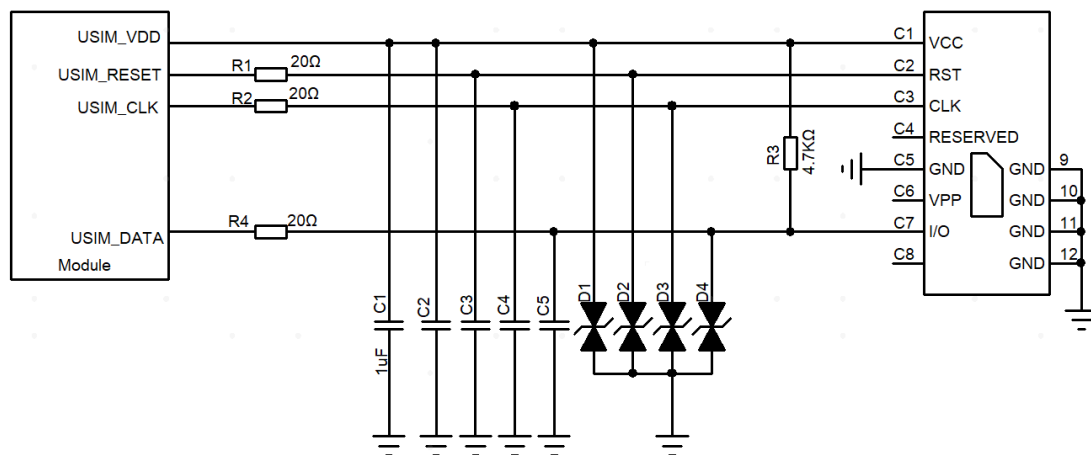
5.3.3 USIM

Signal	Pin SN	I/O	Function description	Remarks
USIM1_DATA	11	B	USIM1 data	This pin is pulled up to USIM1_VDD via a 4.7 kΩ resistor.
USIM1_RESET	12	DO	USIM1 reset	
USIM1_CLK	13	DO	USIM1 clock	
USIM1_VDD	14	PO	USIM1 power	Both 1.8 V and 3.0 V SIM types

				are supported: activation and deactivation with an automatic voltage switch from 1.8 V to 3 V is implemented.
USIM2_VDD	65	PO	USIM2 power	Both 1.8 V and 3.0 V SIM types are supported: activation and deactivation with an automatic voltage switch from 1.8 V to 3.0 V is implemented.
USIM2_RESET	63	DO	USIM2 reset	
USIM2_CLK	62	DO	USIM2 clock	-
USIM2_DATA	64	B	USIM2 data	This pin is pulled up to USIM2_VDD via a 4.7 kΩ resistor.

N706-EA provides two USIM card interfaces and supports dual-SIM dual-standby single-active (contact Neoway FAEs to select the corresponding version, if required). USIM1 is used first by default. Both 1.8 V and 3.0 V SIM types are supported. The reference schematic diagram of the USIM card interface is shown in Figure 5-23.

Figure 5-23 Reference design for USIM card interface (without hot-swap)



Schematic Design Guidelines

- USIM_VDD is the power supply of the USIM card, with load capacity of up to 50 mA. It is solely used for power supply for USIM card; do not power other loads. The pin must be reserved for a regulator capacitor and a filtering capacitor, with a recommended regulator capacitor (C1) of more than 1 μF and a recommended filtering capacitor (C2) of 0.1 μF.
- The USIM_DATA pin is not pulled up internally, so connecting it through a 4.7 kΩ external pull-up resistor to USIM_VCC is required.

- In applications with complex electromagnetic environments that have high requirements for ESD protection, it is recommended to add ESD protection diodes (junction capacitance $\leq 7\text{pF}$) on each signal trace or add an integrated TVS array.
- Connect a resistor ($\leq 20\ \Omega$) to USIM_DATA, USIM_RESET, USIM_CLK, and USIM_DET (close to the card connector) in series to enhance ESD performance.
- C3-C5 are in parallel to each USIM signal trace and are used to mount high-frequency filter capacitors. Their capacitance value should be not more than 33 pF. Please adjust it according to the actual debugging results.



The USIM1 interface of the module does not support the hot-swap function in principle. If you need to implement the function, Pin#79 of the module needs to be compatible with the SIM card detection function, and a 47 k Ω resistor is needed in series at this pin to pull it up to 1.8 V. Meanwhile, the software must be compatible with the USIM hot-swap detection function.

PCB Design Guidelines:

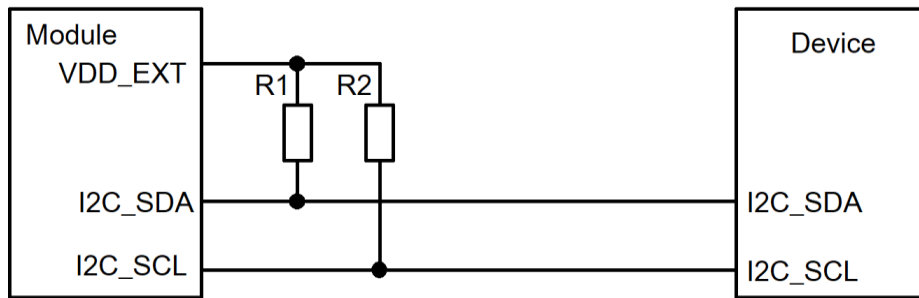
- USIM signals are like to be jammed by RF radiation, resulting in failure to detect the SIM card. Keep the USIM far from the antenna and RF circuit area to prevent jamming.
- The USIM card holder should be placed close to the module and try to ensure the routing length of the USIM card signal traces do not exceed 200 mm.
- On the USIM traces, connect the series resistor and ESD protection component close to the USIM card.
- In order to enhance EMC, surround USIM traces with ground. To prevent USIM_CLK and USIM_DATA from mutually interfering, their routing should not be too close.

5.3.4 I2C

Signal	Pin SN	I/O	Function description	Remarks
I2C2_SDA	66	B	I2C data	Connect this pin to VDD_EXT through a pull-up resistor.
I2C2_SCL	67	DO	I2C clock	
I2C3_SDA	69	B	I2C data	
I2C3_SCL	68	DO	I2C clock	

N706-EA provides two I2C interfaces with 1.8 V level, supporting only the master mode. Their transmission rates in standard mode and fast mode are respectively 100 Kbps and 400 Kbps. For the pull-up resistors, select a proper resistance based on the actual debugging result; it is recommended to select the resistance ranging from 1.8 k Ω to 10 k Ω . Figure 5-24 shows the reference design of the I2C interface.

Figure 5-24 Reference design of I2C interface



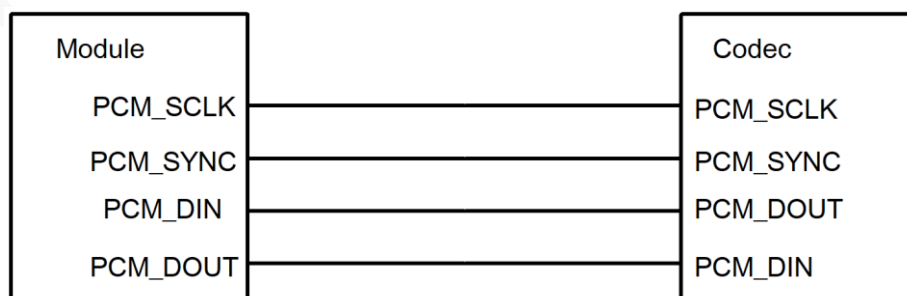
5.4 Audio Interfaces

N706-EA provides one 1.8 V digit PCM interface, which can be multiplexed into PCM interfaces.

Signal	Pin SN	I/O	Function description	Remarks
PCM_CLK	30	DO	PCM data clock	Can be multiplexed into I2S_BCK
PCM_SYNC	31	DO	PCM data frame synchronization	Can be multiplexed into I2S_LRCK
PCM_DIN	32	DI	PCM data input	Can be multiplexed into I2S_SDAT_I
PCM_DOUT	33	DO	PCM data output	Can be multiplexed into I2S_SDAT_O
I2S_MCLK	26	DO	Data clock	-

The following diagram illustrates the PCM connection.

Figure 5-25 Reference design of the PCM chip



Schematic Design Guidelines

- If the logic levels between the module and Codec do not match, you should add a voltage-level translation solution.

PCB Design Guidelines:

- Avoid crossing of PCM signal traces with other traces. If the crossings are unavoidable, 90° crossings should be used to minimize coupling of the traces.
- Route PCM traces as far away as possible from areas where ESD may be introduced.
- Shield I2S_MCLK traces with multi-layer ground during routing.

5.5 Video Interfaces

5.5.1 LCD

N706-EA provides an LCD-dedicated SPI interface, supporting frame rates up to QVGA @ 30fps for LCD display. The following shows the LCD interface pin description.

Table 5-2 LCD interface descriptions

Signal	Pin SN	I/O	Function description	Remarks
LCD_SDC	51	DO	SPI data or command control	
LCD_SIO	50	B	SPI data input/output	
LCD_CS	52	DO	SPI chip select signal of the slave device	
LCD_CLK	53	DO	SPI clock	
LCD_RST	49	DO	LCD reset	
LCD_TE	78	DO	LCD data frame synchronization	-
ISINK	102	PI	LCD sink current input pin, used to adjust backlight brightness.	Connect this pin to the cathode of an LCD and its brightness is adjusted by a sink current. I _{max} = 81 mA.

5.5.2 Camera

N706-EA provides a camera-dedicated SPI interface, supporting frame rates up to 0.3M@24fps for cameras. The following shows the camera interface description.

Table 5-3 Camera interface description

Signal	Pin SN	I/O	Function description	Remarks
CAM_VDD	8	PO	CAM analog power	This analog power is disabled by default. If enabled, the default output is 1.8 V / 100 mA.
CAM_PWDN	81	DO	CAM power-down control	-
CAM_DATA0	55	B	CAM SPI data 0	Camera-dedicated SPI interface
CAM_DATA1	56	B	CAM SPI data 1	
CAM_SPI_CLK	80	DO	CAM SPI clock	
CAM_RST_N	103	DO	CAM reset	-
CAM_MCLK	54	DO	CAM clock signal	-
CAM_I2C1_SCL	57	DO	CAM I2C clock	-
CAM_I2C1_SDA	58	B	CAM I2C data	-

Schematic Design Guidelines

- Notably, CAM_VDD is an LDO output ranging from 1.61 V to 3.2 V (1.8 V by default), which is used to supply power cameras, with a maximum load capacity of 100 mA. Do Not use it for other loads. A filter capacitor must be reserved on the pin in design.

5.6 Other Functional Interfaces

5.6.1 ADC

The module provides two 12-bit ADC channels, with measurable voltage range 0.1 V - VBAT, which can be used for temperature measurement and other related measurement. For details, see the *Neoway_N706_AT_Commands_Manual*.

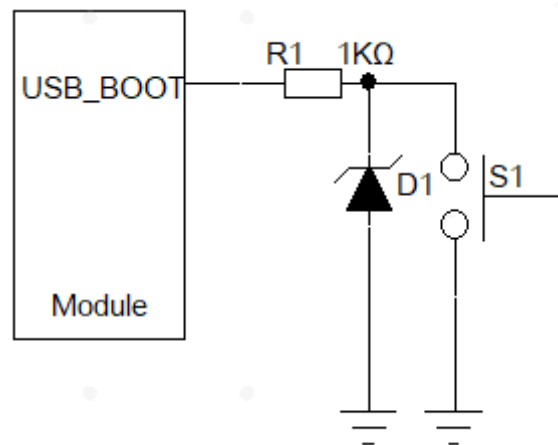
Signal	Pin SN	I/O	Function description	Remarks
ADC0	9	AI	General A/D interface	-
ADC1	96	AI	General A/D interface	-

5.6.2 USB_BOOT

Signal	Pin SN	I/O	Function description	Remarks
USB_BOOT	82	B	Emergency download control	-

The USB_BOOT pin is designed for enabling emergency download mode. To enter forcible download mode you should connect USB_BOOT of the module to ground and then power up the module. This is the last method to handle issues that result in startup or running failures. To facilitate the subsequent software upgrade and debugging of the product, it is recommended to reserve this pin. The following figure shows the reference design of this pin. Adding an ESD component to protect USB_BOOT in the circuit is required.

Figure 5-26 Reference design for the emergency download pin



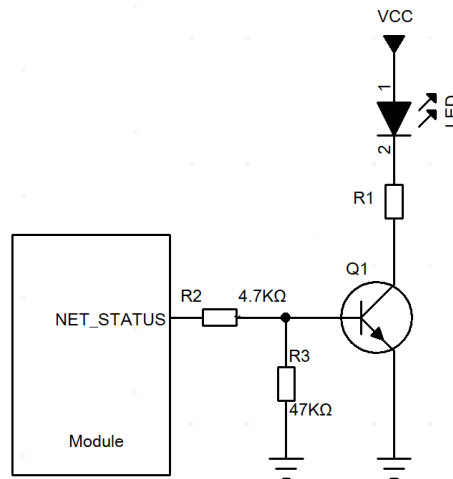
5.6.3 NET_STATUS

Signal	Pin SN	I/O	Function description	Remarks
NET_STATUS	16	DO	Network status indication	Leave this pin open if unused.

NET_STATUS is the network status indicator pin of the module. It outputs PWM waves of duty cycle varying with the status of the module and drives an LED indicator to blink at different modulation frequencies. You can send AT commands to enable the LED indicator to blink in different states. For details, see *Neoway_N706_AT_Commands_Manual*.

The NET_STATUS indication pin outputs a high level of 1.8 V. Do not use this pin to directly drive LED indicator. It is recommended to drive LED indicators through a triode.

Figure 5-27 Driving LED indicators with a triode



5.6.4 STATUS

Signal	Pin SN	I/O	Function description	Remarks
STATUS	25	DO	Operating status indication	Leave this pin open if unused.

STATUS is a network status indicator pin of the module. It outputs PMW waves of duty cycle varying with the status of the module and drives LED indicators to blink at different modulation frequencies. You can send AT commands to enable the LED indicator to blink in different states. For details, see *Neoway_N706_AT_Commands_Manual*.

The STATUS indication pin outputs a high level of 1.8 V, and do not use this pin to directly drive LED indicator. It is recommended to drive LED indicators by controlling a triode. Figure 5-27 shows the recommended reference design.

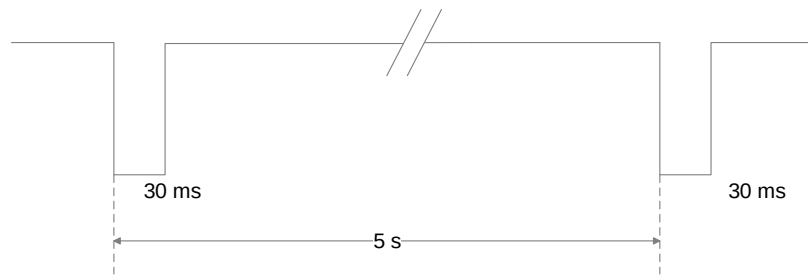
5.6.5 RING

Signal	Pin SN	I/O	Function description	Remarks
RING	20	DO	Incoming call & SMS indicator control	The duration of the low pulse can be configured via software.

- Incoming call RING indication

When there is a voice call, during the ringing process, the RING pin outputs a low pulse with a period of 5s and a pulse width of 30ms. The duration of the low pulse can be configured by software. After the call is answered, the pin restores to high-level output, as shown in the following figure.

Figure 5-28 Incoming call RING indication



- SMS RING indication

When an SMS message arrives, the RING pin will generate a low pulse as a notification. As shown in the following figure.

Figure 5-29 SMS RING indication



5.7 RF Interfaces

Signal	Pin SN	I/O	Function description	Remarks
ANT_GNSS	2	AI	GNSS antenna	
ANT_WIFI_SCAN	98	AI	2.4 GHz WIFI_SCAN antenna	50Ω impedance control.
ANT_MAIN	35	AIO	Main antenna	

5.7.1 ANT_MAIN Antenna Interface

For the module to be applicable to your PCB, the characteristic impedance of the antenna interface should be controlled at 50 Ω and the impedance of the trace from the module's antenna interface to the antenna needs to be kept at 50 Ω to allow reception of radio frequency (RF) signals. In the circuit design, a matching network is essential for antenna matching in the circuit design. The matching network is generally divided into three types: L type, T type, and π type, which are shown in the following figure. The π -type matching circuitry is preferred.

Figure 5-30 L-type RF matching schematic

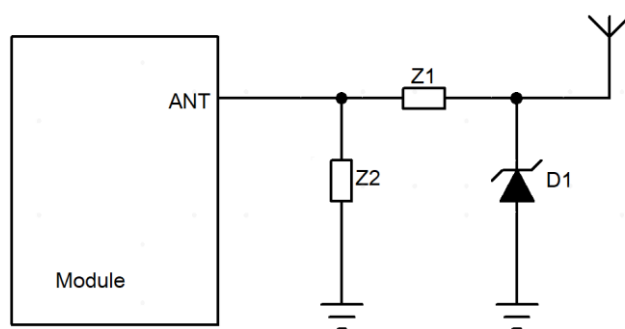
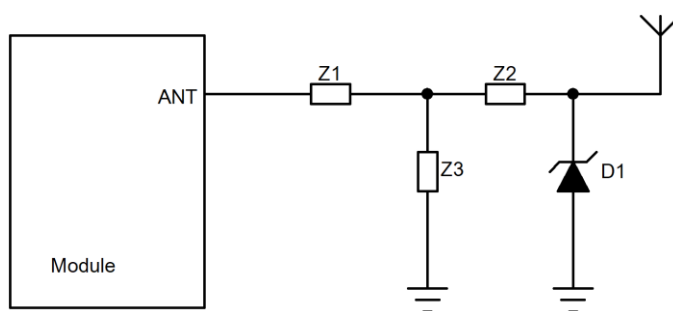
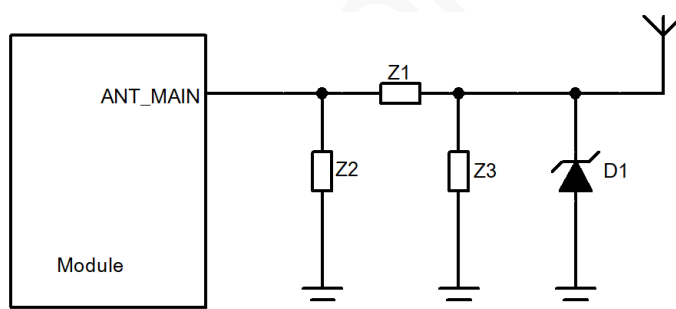


Figure 5-31 T-type RF matching schematic

Figure 5-32 π -type RF matching schematic

Schematic Design Guidelines

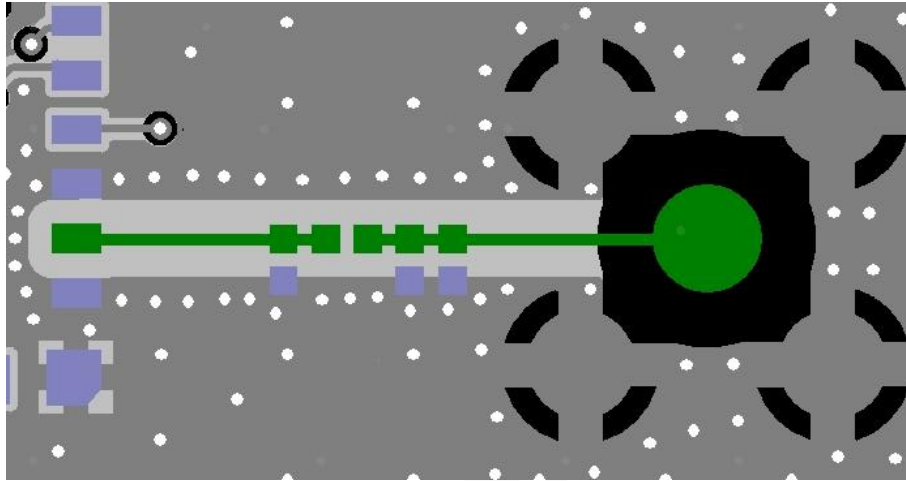
- The components in the RF matching circuit consist of capacitors, inductors, and 0 Ω resistors. Place these RLC components as close to the antenna interface as possible.
- If static electricity may be introduced through the antenna, it is recommended to add ESD components with ultra-low junction capacitance for static electricity protection. TVS diodes with junction capacitance less than 0.5 pF are preferred. Besides, it is necessary to ensure that the reverse breakdown voltage of TVS diodes is greater than 10 V, and TVS diodes with reverse breakdown voltage above 15 V are preferred.

PCB Design Guidelines

- Surround RF traces with grounded copper foil. Drill as many via holes as possible on the copper and control RF trace impedance to 50 Ω .

- Keep the PCB routing between the RF pin and the antenna interface under 50 Ω impedance control, and make the trace length as short as possible.
- To reduce the potential impact on antenna performance caused by the significant parasitic capacitance from the large RF pad in the SMA RF seat, the entire bottom area of the SMA RF seat's RF signal is hollowed out. Additionally, the ground pad next to the module's RF pad is not treated with a thermal pad but should be in complete contact with the ground, as illustrated in the figure below.

Figure 5-33 Recommended RF PCB design



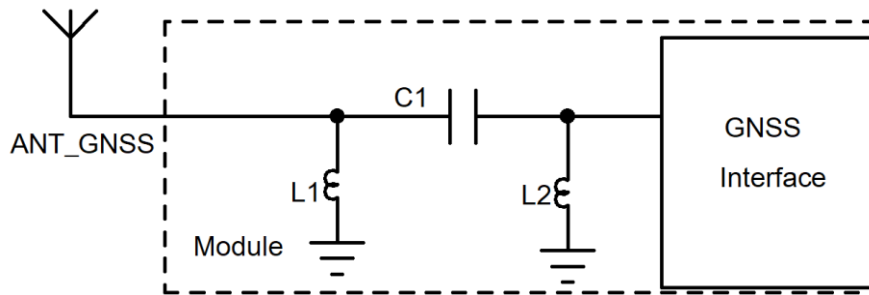
- A reasonable distance should be kept among ANT_MAIN, ANT_WIFI_SCAN, and ANT_GNSS to avoid mutual interference that may affect reception performance.
- On the PCB, keep the RF signals and components far away from digital circuits, switching power supplies, power transformers, power inductors, clock signals, etc.

5.7.2 ANT_GNSS Antenna Interface

GNSS Impedance Control

ANT_GNSS is the GNSS RF interface of the module, which requires a characteristic impedance of 50 Ω . The following shows the GNSS structure inside the module.

Figure 5-34 GNSS RF structure



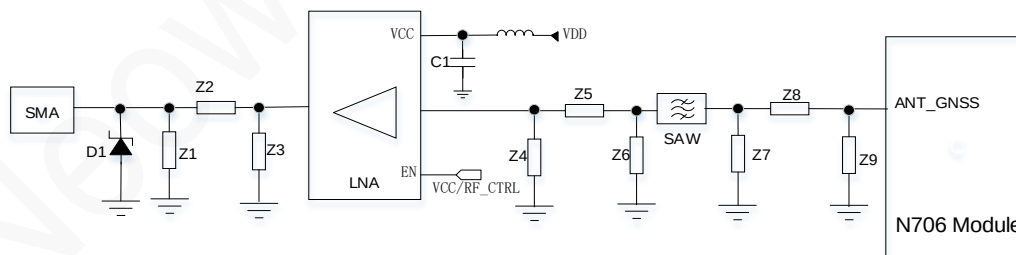
Schematic Design Guidelines

- For matching circuitry between the module interface and GNSS antenna, please refer to the schematic design in section 0
- ANT_MAIN Antenna Interface.
- Add LNAs close to the feeder because the module does not contain one internally. The module's matrix keyboard pins can be used to enable external LNA. For specific pin selection and driver development, contact Neoway FAEs.

Reference design of passive GNSS antenna

After the GNSS antenna receives the GNSS satellite signal, it is transmitted to the ANT_GNSS pin of the N706-EA module through PCB traces. The following figure shows a reference design of the passive GNSS antenna.

Figure 5-35 Reference design of passive GNSS antenna



PCB Design Guidelines:

- External LNA enablement can be directly connected to the LNA's VCC pin or controlled separately using pins #104 and #105. Other GPIO pins are prohibited. Specific driver development needs to be confirmed with Neoway's FAE.

- For cautions of PCB design between GNSS interface and antenna, refer to the PCB design guidelines in section 0
- ANT_MAIN Antenna Interface.
- Keep the GNSS RF circuit away from the main and WIFI_SCAN antenna circuits as far as possible on your application PCB. Otherwise, the RF performance may be affected.

Reference design of active GNSS antenna

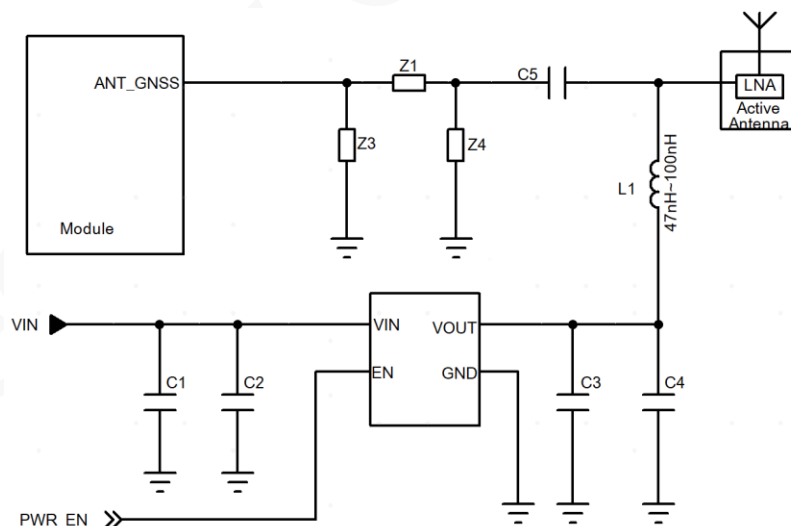


The GNSS chip has a maximum gain limit for the external reception chain. Exceeding this limit can degrade GNSS performance or even damage the chip. If the module has a built-in LNA, the gain requirements for the active antenna are as follows:

The active antenna gain should not exceed 30 dB, with a recommended gain range of 16–19 dB. The noise figure should be less than or equal to 3 dB.

After the GNSS antenna receives GNSS satellite signals, the signals are amplified by the front LNA (Low Noise Amplifier) inside the active antenna and then sent to the ANT_GNSS pin of the module through the feeder and PCB traces. The following figure shows the reference design of the active GNSS antenna.

Figure 5-36 Reference design of active GNSS antenna



PCB Design Guidelines:

- For cautions of PCB design between GNSS interface and antenna, refer to the PCB design guidelines in section 0

- ANT_MAIN Antenna Interface.
- 50 Ω impedance control is required for both antenna feeders and the PCB traces between the antenna interface and LNA, and the traces should not be too long. The power supply of the active antenna is fed via an inductor of 47 nH - 100 nH from the signal line of the antenna.
- The output voltage of the LDO used to supply power to the active antenna is related to the selection of the LNA. You can select the appropriate LDO to supply power to the antenna according to the selected LNA device. As shown in the above figure.
- Keep the GNSS RF circuit away from the main and WIFI_SCAN antenna circuits as far as possible on your application PCB. Otherwise, the RF performance may be affected.

5.7.3 ANT_WIFI_SCAN Antenna Interface

The WIFI_SCAN RF interface only supports the 2.4 GHz frequency band, and its impedance characteristic is required to be 50 Ω .

For the schematic diagram design and PCB design of WIFI_SCAN antenna interface, please refer to the requirements in section 0

ANT_MAIN Antenna Interface.

5.7.4 Antenna Assembly

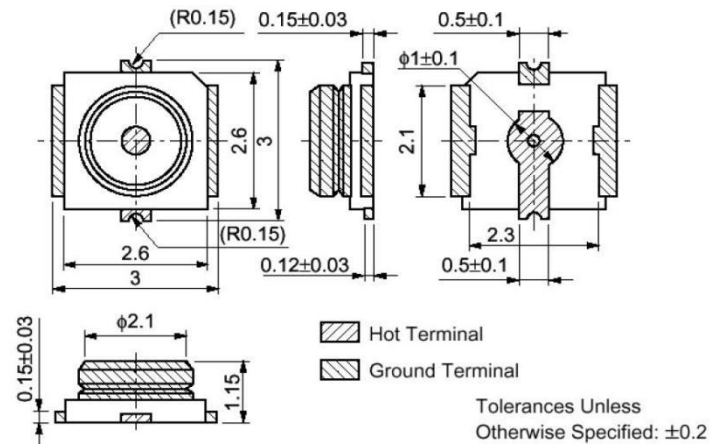
The antenna used by the module must comply with mobile device standards, with a standing wave ratio between 1.1 and 1.5, an input impedance of 50 Ω , and varying gain requirements based on the environment. You can choose an appropriate antenna according to specific application scenarios and environments.

The module antenna interface can connect to rubber rod, sucker, or internal PIFA antennas. Ensure proper shielding between external antennas and RF pins. For RF cable connections, keep the external RF cables away from all interference sources, especially digital signals and switching power supplies.

The following methods are commonly used to assemble antenna:

- Connecting to a GSC RF connector
Recommended model MM9329-2700RA1. The following figure shows its encapsulation specifications. The connection to the external antenna can be made through RF lines.

Figure 5-37 Murata RF connector encapsulation specifications



- The RF cable is connected to the module by means of soldering. However, this method has the stability, consistency, and RF performance degradation issues, and therefore is not recommended.

The following show the two types of connections.

Figure 5-38 RF cable connections

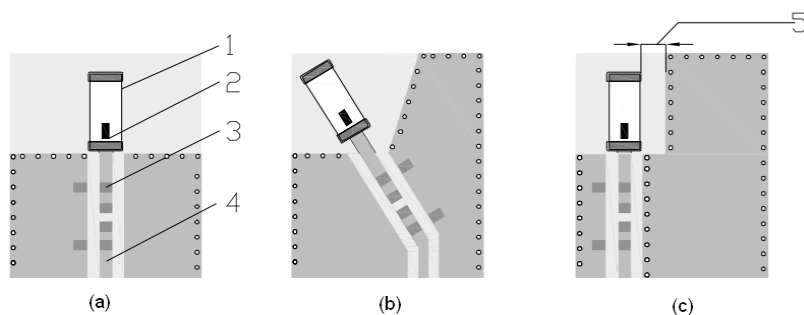


- PCB antenna or chip antenna

The module works in a wide frequency range, but it is difficult for PCB antennas or ceramic antennas to cover a wide frequency. Therefore, this connection method is recommended only for 2.4 GHz Wi-Fi or BT/ BLE antennas.

The following figure shows the layout of the 2.4 GHz ceramic chip antenna. SLDA52-2R540G-S1TF is used as an example.

Figure 5-39 Antenna layout

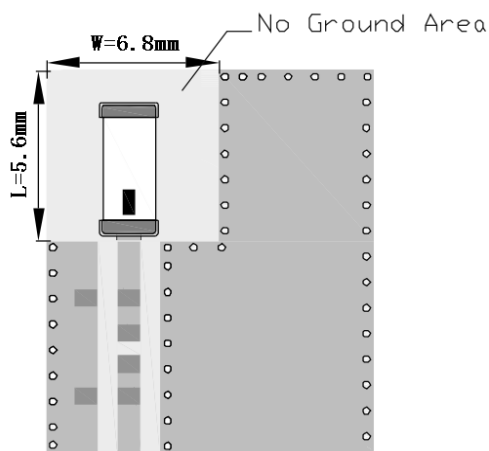


If your PCB is large enough, you can adopt the layout shown in Figure 5-39 (a).

- 1 Chip antenna
- 2 Feeding mark
- 3 Layout pad of the matching circuit
- 4 50 Ω characteristic impedance RF trace

In Figure 5-39, the position marked as "5" depicts the area between the antenna and the ground. As shown in Figure 5-40, the "No Ground Area".

Figure 5-40 Clearance area (No Ground Area) around the antenna on the PCB board



For more details, refer to the antenna manual and instruction documentation.

6 Electrical Characteristics and Reliability

This chapter describes the electrical characteristics and reliability of the N706-EA module, including the input and output voltage and current of the power supply, current consumption of the module in different states, operating and storage temperature range, and ESD protection characteristics.

6.1 Electrical Characteristics



If the voltage is lower than the threshold, the module might fail to start. If the voltage is higher than threshold or there is a voltage burst during the startup, the module might be damaged permanently.

Table 6-1 N706-EA electrical characteristics

Parameter	Minimum value	Typical value	Max. value	Remarks
V_{in}	3.4 V	3.8 V	4.2 V	-
VBAT				The above data was obtained using the Neoway EVB in a laboratory setting. It was measured while connected to a CMW500, with the module operating at its maximum transmission power and a VBAT of 3.8 V.
I_{in}	N/A	N/A	2 A	

Table 6-2 N706-EA current consumption (typical)

Status	Sleep (mA)	Idle (mA)	Active (mA)
Network standard and band			
FDD-LTE: B1, B3, B5, B7, B8, B20, B28	<1.8	<11	<600

Table 6-3 N706-EA-GNSS current consumption (typical)

Description	Condition	Current (mA)
Clod start (AT+CFUN= 0)	Instrument	TBD
Tracking mode (AT+CFUN=0)	Instrument	TBD



The above data was obtained using the Neoway EVB in a laboratory setting. It was measured while connected to a CMW500.

6.2 Temperature Characteristics

Parameter	Minimum value	Typical value	Max. value
Operating	-30°C	25°C	75°C
Extended*	-40°C*	25°C	85°C*
Storage	-40°C	25°C	90°C



In the event that the temperature in the operating environment surpasses the designated thresholds, certain RF performance indicators of the module may deteriorate and fall short of the 3GPP specification requirements. However, despite this, the module's regular functionality will not be significantly affected. It is possible for the RF performance to return to the required 3GPP standards once the temperature is normalized.

6.3 ESD Protection Characteristics

Since electronic products need to undergo strict ESD testing, the module pins need to have ESD protection capabilities. The ESD protection characteristics of the main pins are shown in Table 6-3. You need to add corresponding ESD protection according to the application industry of the product when designing related products to ensure product quality.

Humidity 45% Temperature 25°C

Table 6-4 ESD Protection Characteristics

Contact point	Contact discharge	Air discharge
GND	±8 kV	±15 kV
ANT	±4 kV	±8 kV
Shielding cover	±8 kV	±15 kV

7 RF Characteristics

N706-EA supports FDD-LTE (Cat 1) network mode. This chapter introduces the wireless RF characteristics of N706-EA.

7.1 Operating Frequency Bands

Table 7-1 N706-EA operating frequency bands

Operating frequency band	Uplink	Downlink
FDD-LTE B1	1920 - 1980 MHz	2110 - 2170 MHz
FDD-LTE B3	1710 - 1785 MHz	1805 - 1880 MHz
FDD-LTE B5	824 - 849 MHz	869 - 894 MHz
FDD-LTE B7	2500 - 2570 MHz	2620 - 2690 MHz
FDD-LTE B8	880 - 915 MHz	925 - 960 MHz
FDD-LTE B20	832 - 862 MHz	791 - 821 MHz
FDD-LTE B28	703 - 748 MHz	758 - 803 MHz

7.2 TX Power and RX Sensitivity

Table 7-2 N706-EA RF transmitting power

Band	Max Power	Min. power
FDD-LTE B1	23 dBm $\pm$ 2 dB	< -40 dBm
FDD-LTE B3	23 dBm $\pm$ 2 dB	< -40 dBm
FDD-LTE B5	23 dBm $\pm$ 2 dB	< -40 dBm
FDD-LTE B7	23 dBm $\pm$ 2 dB	< -40 dBm
FDD-LTE B8	23 dBm $\pm$ 2 dB	< -40 dBm
FDD-LTE B20	23 dBm $\pm$ 2 dB	< -40 dBm
FDD-LTE B28	23 dBm $\pm$ 2 dB	< -40 dBm

Table 7-3 N706-EA RF RX sensitivity

Band	RX sensitivity
FDD-LTE B1	≤ -98 dBm
FDD-LTE B3	≤ -98 dBm
FDD-LTE B5	≤ -99 dBm
FDD-LTE B7	≤ -97 dBm
FDD-LTE B8	< -99 dBm
FDD-LTE B20	≤ -97 dBm
FDD-LTE B28	≤ -97 dBm



The preceding indicators are tested in a shielded environment in a laboratory. The LTE band indicators are the test results when the bandwidth is 10 MHz, the modulation mode is QPST and RB is set according to the protocol. On no-shielded environments, deviations may exist in the receiver sensitivity of some individual bands due to the interference.

7.3 GNSS Technical Parameters

Table 7-4 GNSS technical parameters

Parameter	Notice
GPS L1 operating frequency	1575.42±1.023 MHz
GLONASS operating frequency	1597.5 - 1605.9 MHz
BDS operating frequency	1559.1 - 1563.1 MHz
Tracking sensitivity	≤ -159 dBm
Acquisition sensitivity	≤ -148 dBm
Positioning accuracy (open space)	3m@95% (normal temperature)
Hot start (open space)	≤ 4 s (normal temperature)
Cold start (open space)	≤ 35 s (normal temperature)
Update frequency	1Hz
Max. positioning speed	380km/h
CNRin/CNRout	NF ≈ 3.46 dB
GNSS data type	NMEA-0183
GNSS antenna type	Passive/active antenna



The tracking sensitivity, and recapture sensitivity are obtained from the signaling test on GSS7000. The values are the maximum values obtained from multiple measurements performed on samples. External LNA was used during the test.

7.4 WIFI_SCAN Technical Parameters

Table 7-5 WIFI_SCAN TX power and RX sensitivity

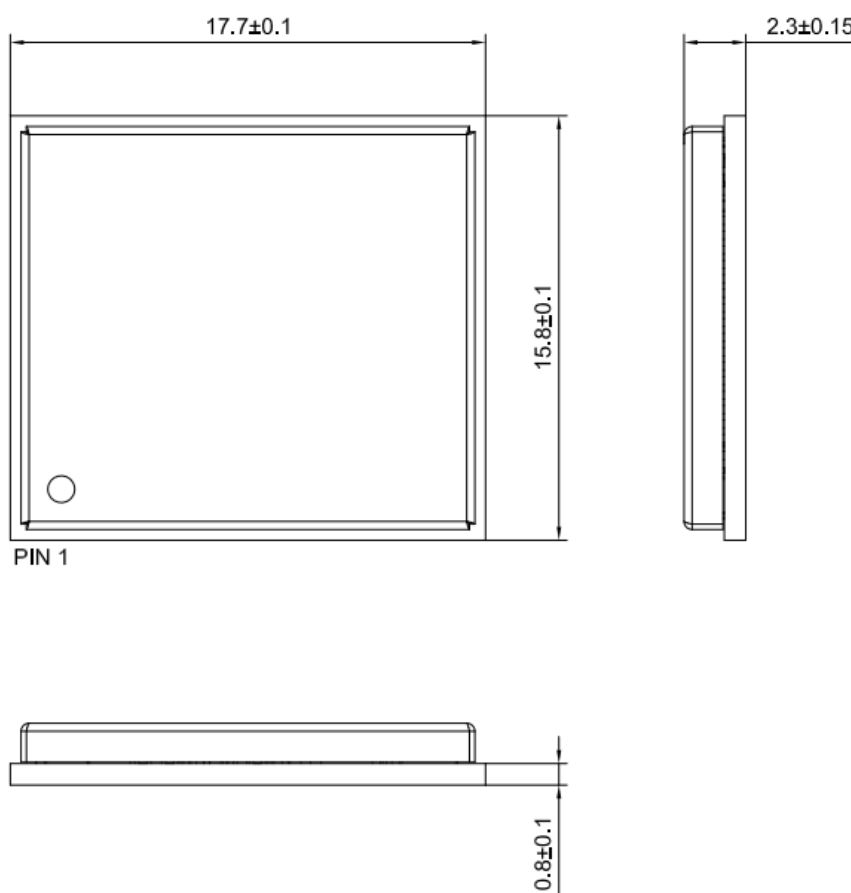
Operating frequency band	Transmitting rate	Transmit power	Receiving sensitivity
802.11b (2.4G)	1/2/5.5/11 Mbps	N/A	-98 dBm

8 Mechanical Characteristics

This chapter describes mechanical characteristics of the N706-EA module.

8.1 Dimensions

Figure 8-1 Top and side view dimensions (unit: mm)



8.2 Labeling

The N706-EA label is laser etched, and can withstand a high temperature of 260°C.

Figure 8-2 N706-EA label



The preceding label is for reference only, the actual effect is subject to the actual product.

8.3 Packaging

Due to the use of surface-mount technology (SMT) for solder reflow in our products, moisture-proof packaging methods are employed to prevent moisture absorption from production to customer use. These methods include aluminum foil bags, desiccants, humidity indicator cards, reels, and vacuum sealing. Such measures ensure the products remain dry, thereby extending their usability.

8.3.1 Tape&Reel

Neoway modules are delivered as reeled tapes as shown below:



8.3.2 Moisture Sensitivity Level



N706-EA is a level 3 moisture-sensitive electronic element, in compliance IPC/ JEDEC J-STD-020 standard..

Production environment condition: 30°C/60%

The module should be pre-baked under the following circumstances:

- The vacuum-sealed packaging is removed for more than 48 hours.
- The relative humidity is greater than 10% (you can see the humidity card that comes with the package).

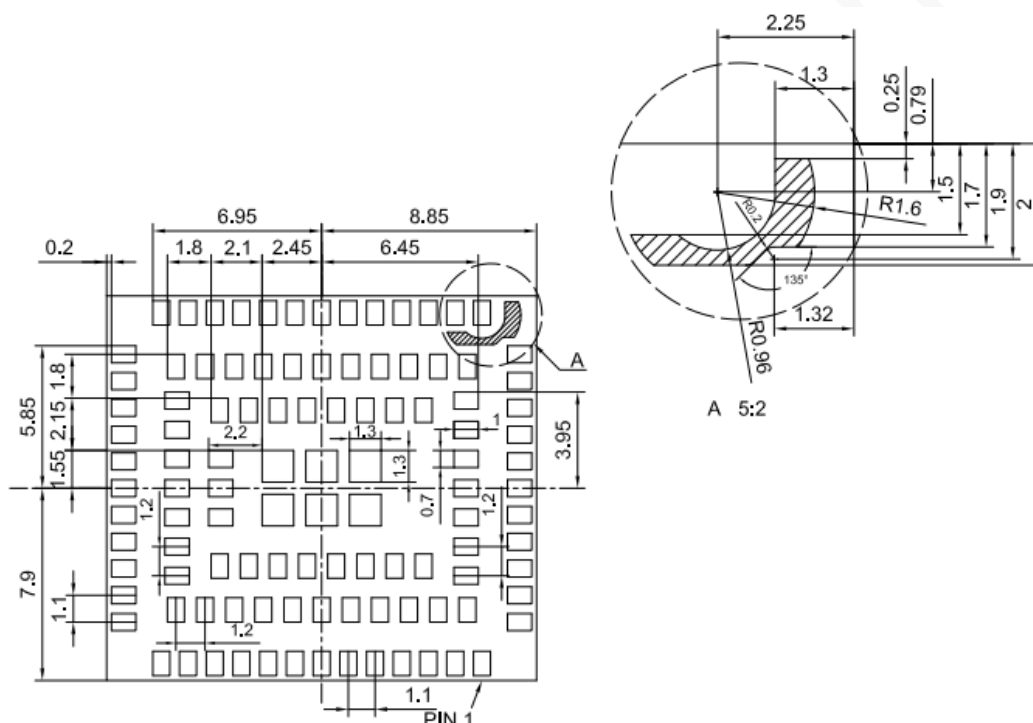
After the module is unpacked, if it is exposed to air for a long time, the module will get damped, and may be damaged during reflow soldering or laboratory soldering. Bake it before mounting the module. The baking conditions depend on the moisture degree. It is recommended to bake the module at temperatures higher than 120 degrees for more than 6 hours. Do not bake Neoway modules while contained in a tape and rolled up in reels. For baking, place modules individually onto the oven tray.

9 Mounting N706-EA onto Application Board

This chapter describes the package of N706-EA, the recommended footprint of the application PCB, and SMT specifications.

9.1 PCB Package

Figure 9-1 Bottom view of N706-EA PCB package (unit: mm)



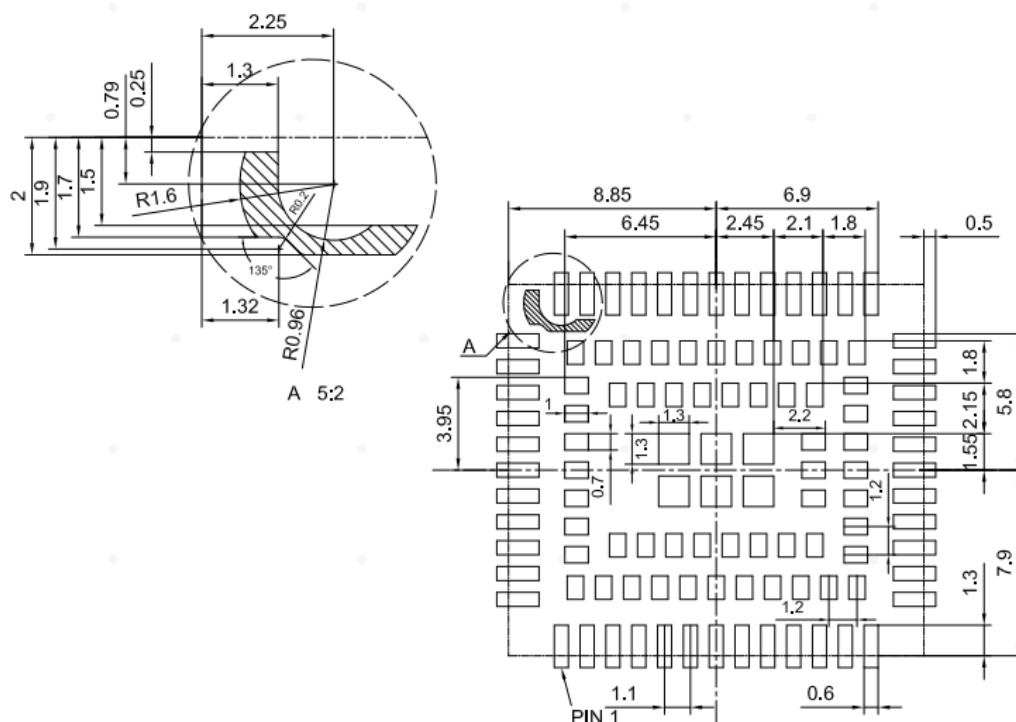
9.2 Application PCB Package

The N706-EA module has a total of 109 pins in LGA package, and the recommended application PCB package is as follows:



- To ensure proper operation of the module, only GND via-holes and copper pouring are allowed within the shaded area "B" of the PCB package.
- To achieve higher yield during module production, it is recommended that the distance between other components on the PCB board and the module pads be at least 3 mm to avoid the risk of tin connection when using stepped stencil.

Figure 9-2 N706-EA PCB foot print (top view, unit: mm)



9.3 Stencil

The recommended stencil thickness is at least 0.15 mm to 0.20 mm. For detailed information on stencil aperture design, please refer to the *Neoway_Communication_Module_SMT_Application_Guide*.

9.4 Solder Paste

The thickness of solder paste and the flatness of PCB are essential for the production yield.

Neoway modules use lead-free solder paste technology. It is recommended to use the same type of solder paste as used in our module production technique.

Users are generally not recommended to use leaded solder paste due to the following reason:

Leaded solder paste has a melting point 35°C lower than lead-free, resulting in lower temperatures and shorter times in reflow processes, which may cause LCC/LGA components in the module to be in a semi-melted state during the second reflow, leading to cold solder joints.

When using solder pastes with lead, ensure the reflow temperature exceeds 220°C for over 45 seconds, with a peak temperature of 240°C.

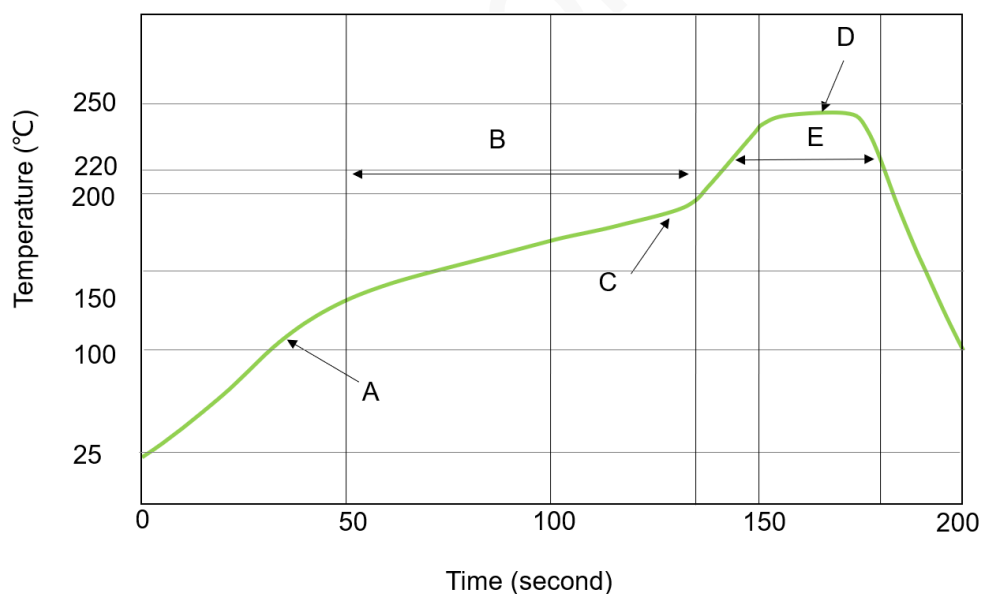
9.5 Oven Temperature Profile



Due to abnormal temperatures, thermal sensitive devices may fail and cause other adverse effects. Neoway assumes no responsibility for such consequences.

If the PCB is large, it is important to avoid bending of the printed circuit material during an SMT process. So a bending prevention tool must be placed on the bottom of the printed circuit board. It is recommended to use loading tools during the SMT and reflow soldering process to avoid poor solder joint caused by PCB bending.

Figure 9-3 Oven temperature profile



Technical parameters:

- Ramp up rate: 1 to 3°C/sec
- Ramp down rate: -3 to -1°C/sec
- Soaking zone: 150 - 180°C, Time: 60 - 100s

- Reflow zone: >220°C, Time: 40 - 90s
- Peak temperature: 235 – 245°C

For information about cautions in N706-EA storage and mounting, refer to *Neoway_Reflow_Soldering_Guidelines_For_Surface-Mounted_Modules*.

When manually removing a module from your application's PCB board, it is important to use heat guns with large openings. Adjust the temperature to about 245°C (depending on the type of solder paste used) and heat the module until the solder paste melts. Use tweezers to remove the module, being careful not to shake it at high temperatures. Shaking the module may cause components inside to become misplaced.

A Abbreviations

Abbreviation	Full name
ADC	Analog-to-Digital Converter
AI	Analog Input
AO	Analog Output
AIO	Analog Input/Output
ARM	Advanced RISC Machine
bps	Bits per Second
CCC	China Compulsory Certification
CEP	Circular Error Probable
CNR	Carrier to Noise Rate
CS	Chip Select
CTS	Clear to Send
DC	Direct Current
DCS	Digital Cellular System
DI	Digital Input
DIO	Digital Input/Output
DL	Downlink
DO	Digital Output
DPSK	Differential Phase Shift Keying
DQPSK	Differential Quadrature Phase Shift Keying
DRX	Discontinuous Reception
DTR	Data Terminal Ready
ECM	Ethernet Control Model
eDRX	Extended DRX
EGSM	Enhanced GSM
ESD	Electronic Static Discharge
ESR	Equivalent Series Resistance
EVK	Evaluation Kit

FDD	Frequency Division Duplexing
FPC	Flexible Printed Circuit
FTP	File Transfer Protocol
GFSK	Gauss Frequency Shift Keying
GLONASS	GLOBAL NAVIGATION SATELLITE SYSTEM
GNSS	Global Navigation Satellite System
GPIO	General Purpose Input Output
3GPP	3rd Generation Partnership Project
GPRS	General Packet Radio Service
GPS	Global Positioning System
GSM	Global System for Mobile Communications
I2C	Inter-Integrated Circuit
IO	Input/Output
ISP	Image Signal Processor
LCC	Leadless Chip Carriers
LCD	Liquid Crystal Display
LED	Light Emitting Diode
LGA	Land Grid Array
LTE	Long Term Evolution
MCLK	Main Clock
MCU	Microcontroller Unit
MIC	Microphone
PCB	Printed Circuit Board
PCS	Personal Communications Service
PWM	Pulse Width Modulation
QVGA	Quarter Video Graphics Array
RAM	Random Access Memory
RF	Radio Frequency
ROM	Read-only Memory
RTC	Real Time Clock
SD	Secure Digital
SDIO	Secure Digital Input Output
SPK	Speaker

SPI	Serial Peripheral Interface
UART	Universal Asynchronous Receiver-Transmitter
UL	Uplink
USB	Universal Serial Bus
USIM	Universal Subscriber Identity Module
VBAT	Battery Voltage
VSWR	Voltage Standing Wave Ratio
Wi-Fi	Wireless Fidelity
WCDMA	Wide-band Code Division Multiple Access
WCI	Wireless Coexistence Interface
WLAN	Wireless Local Area Network